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To Fight or to Give Up? Dynamic Contests with a Deadline

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Abstract. We study dynamic contests between two players whose performance is determined jointly by effort and luck. The players observe each other's positions in real time. There is a fixed deadline, and the player with a higher performance at the deadline wins the contest. We fully characterize the Markov perfect equilibrium for heterogeneous players. Effort is high when the players are tied but collapses quickly when one of them assumes a lead, due to a dynamic momentum effect. Therefore, total expected effort does not necessarily increase in the prize or in the players' abilities. We discuss implications for contest design and propose splitting the contest to cool off competition and introducing optimal head-starts for heterogeneous players as possible solutions.

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Keywords: dynamic contest • finite time horizon • Brownian motion • momentum effect

1. Introduction

Picture yourself working in sales alongside another employee. The company pays a bonus at the end of the month to the one of you who sells more product. You work every day and observe each other's progress in real time. Sales are affected not just by your effort but also by luck, or random factors beyond your control, such as customers arriving at random times. How would your effort change throughout the month? How would you react to being ahead or behind your rival on any given day? Would you work harder or relax as the end of the month approaches?

In this paper, we analyze dynamic contests characterized by three features: (i) there is a fixed deadline, (ii) participants' progress is publicly observable in real time, and (iii) participants' performance is affected jointly by effort and luck. Such contests serve as a model for the situation described previously, as well as for other similar environments, including competition in organizations and political campaigns. We model competition in continuous time, with a stochastic state ("score") evolving as a Brownian motion with a drift proportional to the difference between the players' costly efforts. The players observe the score, and the one who has a lead at the deadline wins the contest and receives a fixed prize.

We fully characterize the Markov perfect equilibrium for players with (possibly) heterogeneous abilities, modeled as different costs of effort. We show that the equilibrium effort is essentially zero outside a "window of opportunity"—a range of scores where differences in performance are sufficiently small. As the contest progresses, this window narrows down,

but competition inside the window intensifies. Unless abilities are very different, when a player is far behind, she has little incentive to exert effort because the likelihood of catching up with the leader is low. The leader's effort then drops as well. As time progresses, catching up becomes even less likely, and the window of opportunity shrinks. Within the window, competition intensifies because the marginal impact of effort on the likelihood of winning increases.

When players are symmetric or nearly symmetric in ability, the main feature of the equilibrium is a dynamic momentum effect: Individual effort is maximized for a player with a positive lead, and the leader's effort always exceeds that of the follower. Intuitively, by increasing effort, the leader further increases her probability of winning and decreases future effort costs; in contrast, the trailing player moves closer to the zone of intense competition and increases her future effort costs.¹ More generally, as long as the noise is not too strong, play in the contest evolves in one of two regimes: (Nearly) free diffusion and intense competition. Free diffusion happens outside the window of opportunity where efforts are close to zero. Intermittent periods of intense competition can happen when the state spontaneously enters the window of opportunity. However, the window acts as a repeller because the drift created by the dynamic momentum effect pushes the state away from it on either side. This instability of the state where the players are tied implies that once a player assumes a lead, it is difficult to reverse. For the leader to change, a large enough fluctuation of noise, or an

unlikely sequence of shocks, should overcome a “barrier” that gets progressively steeper.

As a consequence of these dynamic effects, total expected effort in the contest is nonmonotone in the prize, increasing when the prize is relatively low but decreasing for high prizes. When the prize is high, the players exert a lot of effort initially, and because of noise and the momentum effect one of them quickly assumes a big lead; effort then collapses and is unlikely to ever recover. Increasing both players’ abilities has a similar effect. Thus, in contrast to static contests where higher incentives always lead to more effort, increasing incentives in a dynamic contest does not necessarily generate a higher total effort over time. From a managerial perspective, this implies that cooling off competition may be beneficial. We show that splitting the contest into several shorter subcontests with smaller prizes produces the desired result. Intuitively, splitting the contest works for two reasons. First, having smaller prizes can by itself raise total effort due to effort provision increasing in states away from the tie; second, having the contest restart automatically resets the state to zero where high effort is guaranteed.

When players are sufficiently asymmetric, the stronger player’s effort exceeds the weaker player’s effort in most, and possibly all, states. If the contest starts at zero score, total effort will be low because it is relatively cheap for the stronger player to get ahead, and the weaker player has no incentive to even try to compete. However, significant effort can be generated by introducing a *head-start*—an advantageous initial position for the weaker player.² Optimal head-starts are large enough to provide the weaker player a sufficient fighting chance, but not too large, so that the stronger player is incentivized to work hard to try and close the gap. Thus, high effort can be sustained dynamically throughout the contest. In fact, we show that a contest between a strong player and a weak player with an optimal head-start can generate a higher effort than a contest between two strong players.

1.1. Related Literature

The literature on *contests*—games, in which players compete for a valuable prize by exerting costly effort—is vast. Contest models can be divided into those with perfect and imperfect discrimination, depending on whether luck, or noise, plays a role in the winner determination process. For noisy contests, which are more relevant for this paper,³ most authors use a variant of one of two seminal models by Tullock (1980) and Lazear and Rosen (1981).⁴ Although static models are prevalent, the literature on dynamic contests is also very rich. For our context, we focus the review on models of effort choice in repeated contests with a fixed number of players, leaving aside settings involving entry and elimination,⁵ as well as settings where

players are able to control the riskiness of their actions and noise is endogenous.⁶

In the literature on dynamic contests, two-period models are widely used for tractability reasons (Yildirim 2005, Casas-Arce and Asís Martínez-Jerez 2009, Gershkov and Perry 2009, Aoyagi 2010, Ederer 2010, Goltsman and Mukherjee 2011, Ridlon and Shin 2013). These papers explore various elements of contest design, such as the role of head-starts, interim feedback, handicaps, and bonuses. A two-stage Lazear-Rosen type model is, however, insufficient to see the basic dynamic effects discussed in the opening paragraph of this paper because, for ex ante symmetric players, the equilibrium effort remains symmetric in both stages. Increasing the number of stages beyond two, with public feedback, quickly makes the model intractable.

In “longer” models with nontrivial dynamic effects, each player typically controls an individual stochastic process, such as a Brownian motion or Poissonian arrivals of breakthroughs. These models can be divided into private and public information ones, depending on whether real-time information about the players’ relative positions is revealed.⁷ Examples of private information models include Taylor (1995), Seel and Strack (2013, 2016), Lang, Seel and Strack (2014), Chen, Jiang and Knyazev (2016), and Benkert and Letina (2020).

The most relevant for this paper are models where the players’ relative position is public information throughout the game. Here, our paper differs from (most) others in how the winner of the contest is determined. In discrete time, Konrad and Kovenock (2009) and Ryvkin (2011) study best-of- $(2n - 1)$ games (also known as multibattle contests), in which the player who is the first to win n stage games is the winner overall. A more general approach to discrete-time dynamic contests encompassing both the tug-of-war and best-of- $(2n - 1)$ games using the formalism of finite automata has been developed by Ewerhart and Teichgräber (2019). Moscarini and Smith (2011) and Cao (2014) analyze tug-of-war type contests in continuous time where in order to win, a player must achieve a certain lead over her opponent. A discrete time version of this approach is used by Harris and Vickers (1987).

The closest setup to ours is used by Daley and Wang (2018). Like us, they also consider a contest in continuous time, with a fixed end date T , where each player’s state is a Brownian motion with a drift proportional to effort. However, unlike us, Daley and Wang (2018) assume that the players’ states are private information with the exception of one moment $\tau \leq T$ when the relative position is revealed. They show that there is an intermediate optimal τ maximizing total expected effort. Furthermore, in a chapter of

his PhD dissertation, Wang (2017) looks at a setup similar to ours and also allows for a random device to determine the winner if at time T the distance between the players is within a certain window $\Omega \geq 0$. Although this aspect of the model is more general, the analysis is based entirely on numerical explorations.

The rest of the paper is structured as follows. In Section 2, we present the model setup. The equilibrium is characterized in Section 3, and its properties are discussed in Section 4. Implications for contest design are studied in Section 5, and Section 6 concludes. All missing proofs and derivation details are collected in Appendix A.

2. Model

We consider a contest of two risk-neutral players indexed by $i \in \{1, 2\}$. The contest starts at $t = 0$ and ends at a fixed deadline $t = T > 0$, where time $t \in [0, T]$ is continuous. Player i chooses effort μ_{it} at a flow cost $c_i \mu_{it}^2/2$, where parameters $c_i > 0$ may be heterogeneous and measure the players' abilities, with a lower c_i corresponding to higher ability.

By exerting effort, the players drive the stochastic "score" of the game, X_t , that evolves as

$$dX_t = (\mu_{1t} - \mu_{2t})dt + \sigma dW_t, \quad X_0 = x_0. \quad (1)$$

Here, W_t is the standard Wiener process, $\sigma > 0$ is a measure of noise intensity, and the initial score is x_0 . Player 1 (2) wins the contest if $X_T > (<)0$.⁸ The winner receives a fixed prize $V > 0$, and the loser receives a prize normalized to zero. The players observe the score at all times. We say that player 1 (2) is the leader at t if $X_t > (<)0$, and the other player is called the follower.

Score X_t can be interpreted as the relative performance of the two players on some evaluation scale. For example, employees in an organization can be evaluated periodically and receive "points" accumulating over time, in which case X_t is the difference between the players' point totals. For salespeople, X_t can represent the difference in aggregate sales, for politicians—a difference in the polls, for pharmaceutical companies competing to develop a new treatment—a difference in how successful their clinical trials have been. Noise W_t can represent a genuine luck component—the influence on performance of factors other than effort, beyond the players' control. Noise can also be interpreted as a measurement error, whereby the evaluator cannot accurately *perceive* differences in performance. For example, it may be difficult for managers in an architectural firm or in a pharmaceutical company to evaluate individual employees' contribution to a complex project.

The state of the game with $X_t = x$ is fully characterized by a pair $(x, t) \in \mathbb{R} \times [0, T]$. Let $v_i(x, t)$ denote the optimal value function of player i in state (x, t) . For

player 1, treating the path of player 2's effort, $(\mu_{2\tau})_{\tau \in [t, T]}$, as given,

$$v_1(x, t) = \max_{(\mu_{1\tau})_{\tau \in [t, T]}} \mathbb{E} \left[V \mathbb{1}_{X_T > 0} - \frac{c_1}{2} \int_t^T \mu_{1\tau}^2 d\tau \mid X_t = x \right]. \quad (2)$$

Here, the expectation is taken over all paths $(X_\tau)_{\tau \in [t, T]}$ starting at $X_t = x$ and controlled by the two efforts according to (1). The value function of player 2 is defined similarly, with the indicator replaced by $\mathbb{1}_{X_T < 0}$.

We focus on the Markov perfect equilibrium in which the players choose efforts in state (x, t) according to policy functions $\mu_{it} = m_i(x, t)$ that depend only on the state. The corresponding Hamilton-Jacobi-Bellman (HJB) equation for player 1, derived in Appendix A, has the form

$$0 = \max_{\mu_{1t}} \left[-\frac{c_1 \mu_{1t}^2}{2} + v_{1x}(x, t)(\mu_{1t} - \mu_{2t}) + \frac{\sigma^2}{2} v_{1xx}(x, t) + v_{1t}(x, t) \right]. \quad (3)$$

The equation for player 2 is similar. Under the assumptions $v_{1x}(x, t) > 0$ and $v_{2x}(x, t) < 0$, which we verify later, the unique maxima are reached at $\mu_{1t} = v_{1x}(x, t)/c_1$ and $\mu_{2t} = -v_{2x}(x, t)/c_2$. This gives the system of equations for the value functions,

$$\begin{aligned} \frac{1}{2c_1} v_{1x}(x, t)^2 + \frac{1}{c_2} v_{1x}(x, t) v_{2x}(x, t) + \frac{\sigma^2}{2} v_{1xx}(x, t) + v_{1t}(x, t) &= 0, \\ \frac{1}{2c_2} v_{2x}(x, t)^2 + \frac{1}{c_1} v_{1x}(x, t) v_{2x}(x, t) + \frac{\sigma^2}{2} v_{2xx}(x, t) + v_{2t}(x, t) &= 0. \end{aligned} \quad (4)$$

The partial differential equations (PDEs) (4) are subject to the boundary conditions $v_1(-\infty, t) = 0$, $v_1(+\infty, t) = V$, $v_2(-\infty, t) = V$, and $v_2(+\infty, t) = 0$, and the terminal conditions $v_1(x, T) = V$ for $x > 0$, $v_1(x, T) = 0$ for $x < 0$, $v_2(x, T) = 0$ for $x > 0$, and $v_2(x, T) = V$ for $x < 0$.⁹

3. Characterizing the Equilibrium

Using a standard approach to solving the heat equation, parabolic PDEs (4) can be reduced to ODEs. Consider a composite variable $z = x/\sigma\sqrt{T-t}$, and let $v_i(x, t) = VF_i(z)$. This is convenient because all the boundary and terminal conditions are satisfied automatically if $F_1(-\infty) = 0$, $F_1(+\infty) = 1$, $F_2(-\infty) = 1$, and $F_2(+\infty) = 0$. As will be seen later, $F_1(z)$ and $F_2(-z)$ are increasing functions and thus have the properties of a cumulative distribution function (CDF). Letting $f_i(z) = F'_i(z)$, we obtain for the derivatives of v_i ,

$$\begin{aligned} v_{ix}(x, t) &= \frac{Vf_i(z)}{\sigma\sqrt{T-t}}, \quad v_{ixx}(x, t) = \frac{Vf'_i(z)}{\sigma^2(T-t)}, \\ v_{it}(x, t) &= \frac{Vf_i(z)z}{2(T-t)}. \end{aligned} \quad (5)$$

With $w_i = V/c_i\sigma^2$, Equations (4) then become

$$\begin{aligned} f'_1(z) &= -w_1 f_1(z)^2 - 2w_2 f_1(z)f_2(z) - f_1(z)z, \\ f'_2(z) &= -w_2 f_2(z)^2 - 2w_1 f_1(z)f_2(z) - f_2(z)z. \end{aligned} \quad (6)$$

We will sometimes refer to parameters w_i combining the prize value, costs, and noise intensity as (effective) *abilities* of the two players. As we show later, many comparative statics of the equilibrium can be formulated in terms of these parameters.

It is instructive to compare (6) to the corresponding equation for the derivative of the value function in Moscarini and Smith (2011) who consider a symmetric tug-of-war contest. Setting $w_1 = w_2$ and assuming, from symmetry, that $f_2(z) = -f_1(-z)$, our equations for $f_i(z)$ become similar to theirs, with the exception of the linear term $-f_i(z)z$ on the right-hand side. This term corresponds to v_{it} in the original PDEs (4) and emerges because of nonstationarity. In the stationary setup of Moscarini and Smith (2011), z is the state variable measuring the distance between the players. In our case, the composite variable z is the distance x rescaled by $\sigma\sqrt{T-t}$ —the standard deviation of the noise increment $\sigma(W_T - W_t)$ accumulating between t and T . As time progresses, the scale on which changes in x are consequential is shrinking.

Based on their definitions, functions $f_i(\cdot)$ must satisfy the normalization conditions $\int_{-\infty}^{+\infty} f_{1,2}(z) dz = \pm 1$. The following lemma summarizes two useful properties of $f_1(\cdot)$ and $f_2(\cdot)$.

Lemma 1. *If $(f_1(z), f_2(z))$ is a continuous solution to system (6) that satisfies the normalization conditions, then*

- (i) $f_1(z) > 0$ and $f_2(z) < 0$ for all $z \in \mathbb{R}$; and
- (ii) $f_1(z)$ and $-f_2(z)$ are unimodal and vanish as $z \rightarrow \pm\infty$.

Part (i) of Lemma 1 shows that the contest is *pervasive*, that is, the equilibrium effort of both players is positive in all states. This is because, at zero effort, the marginal cost of effort is zero, but the marginal benefit is positive (see (3)). Moscarini and Smith (2011) and Cao (2014) identify a similar property for tug-of-war contests.

Part (ii) of Lemma 1, together with part (i), implies that the equilibrium effort is single peaked in x . As we show later, effort decays exponentially away from the peak, that is, it is very small outside a window of width proportional to $\sigma\sqrt{T-t}$. Moreover, the window narrows over time. For tug-of-war contests with quadratic costs, Moscarini and Smith (2011) and Cao (2014) similarly show that effort is either unimodal or monotone in the state.¹⁰

As a next step, it is convenient to seek a solution to (6) in the form $f_i(z) = A_i(z)e^{-z^2/2}$. Equations (6) then transform into

$$\begin{aligned} A'_1(z) &= -[w_1 A_1(z)^2 + 2w_2 A_1(z)A_2(z)]e^{-z^2/2}, \\ A'_2(z) &= -[w_2 A_2(z)^2 + 2w_1 A_1(z)A_2(z)]e^{-z^2/2}. \end{aligned} \quad (7)$$

The amplitude functions $A_i(\cdot)$ are sign-definite and must satisfy the normalization conditions $\int_{-\infty}^{+\infty} A_{1,2}(z)e^{-z^2/2} dz = \pm 1$.

To proceed with the solution, define $A_{\pm}(z) = (w_1 A_1(z) \pm w_2 A_2(z))/2$. The original amplitudes can then be written as $A_{1,2}(z) = (A_+(z) \pm A_-(z))/w_{1,2}$. Using (7), functions $A_{\pm}(z)$ and their derivatives can be connected via the following system of first-order ODEs (for details, see Appendix A):

$$\begin{aligned} A'_+(z) &= [A_-(z)^2 - 3A_+(z)^2]e^{-z^2/2}, \\ A'_-(z) &= -2A_+(z)A_-(z)e^{-z^2/2}. \end{aligned} \quad (8)$$

The normalization conditions for $A_{\pm}(z)$ follow from those for $A_{1,2}(z)$:

$$\int_{-\infty}^{+\infty} A_{\pm}(z)e^{-z^2/2} dz = \frac{w_1 \mp w_2}{2}. \quad (9)$$

We will now construct a solution to (8) that satisfies (9). We start with the following lemma providing three properties of such solutions.

Lemma 2. *Suppose $(A_+(z), A_-(z))$ is a continuous solution to (8) satisfying (9). Then*

- (i) $A_-(z) > 0$ for all $z \in \mathbb{R}$;
- (ii) $A_-(z)$ is either strictly monotone or unimodal; and
- (iii) $A_+(z) < (>) 0$ when $A_-(z)$ is increasing (decreasing), and changes sign at most once, at the mode of $A_-(z)$ if one exists.

Recall that the equilibrium efforts are $m_{1,2}(x, t) = \pm v_{1,2x}(x, t)/c_{1,2}$. Equation (5) then gives $m_{1,2}(x, t) = \pm \sigma w_{1,2} f_{1,2}(z)/\sqrt{T-t}$ or, using the definitions of $A_{1,2}(z)$ and $A_{\pm}(z)$,

$$m_{1,2}(x, t) = \frac{\sigma}{\sqrt{T-t}} [A_-(z) \pm A_+(z)]e^{-z^2/2}. \quad (10)$$

Consider now the two cases identified in part (ii) of Lemma 2. If $A_-(z)$ is strictly increasing (decreasing) and hence, from part (iii) of the lemma, $A_+(z)$ is everywhere negative (positive), we have $m_1(x, t) < (>) m_2(x, t)$ for all states $(x, t) \in \mathbb{R} \times [0, T]$. That is, one of the players—the *dominant* player—always exerts a higher effort. We will refer to this case as *weak competition*. As seen from (9), this is only possible when players are heterogeneous (indeed, if $w_1 = w_2$, then $A_+(z)$ has to change sign), although, as we show later, heterogeneity alone is not sufficient.

If $A_-(z)$ is unimodal, with a mode at z_m , it follows from (10) and Lemma 2 that the two efforts are single-crossing, with $m_1(x, t) = m_2(x, t)$ at (x, t) such that $z_m = x/\sigma\sqrt{T-t}$. Using the terminology of Cao (2014), z_m therefore defines a time-dependent *pivot* point $x_m(t) = \sigma z_m \sqrt{T-t}$. We will refer to the case when a pivot exists as *strong competition*. In this case, the region to the right (to the left) of the pivot is player 1's (player 2's) *dominance zone* where the corresponding player's effort is higher.

Next, we fully characterize the solution to (8). Consider function $\gamma : (-1, 1) \rightarrow \mathbb{R}$ defined by

$$\gamma(u) = \frac{u}{1-u^2} + \frac{1}{2} \ln \frac{1+u}{1-u}. \quad (11)$$

It is strictly increasing, odd, with $\gamma(0) = 0$, $\gamma(u) = -\gamma(-u)$, and diverging as $u \rightarrow \pm 1$. Its inverse, $\gamma^{-1}(\cdot)$, therefore, is well defined for all real values of the argument, strictly increasing, odd, and satisfies $|\gamma^{-1}(\cdot)| < 1$. Furthermore, define constants

$$\rho_1 = \frac{e^{w_1} + e^{-w_2} - 2}{e^{w_1} - e^{-w_2}}, \quad \rho_2 = \frac{e^{w_2} + e^{-w_1} - 2}{e^{w_2} - e^{-w_1}}. \quad (12)$$

It is easy to check that $\rho_{1,2} \in (-1, 1)$, $\rho_1 + \rho_2 > 0$, and $\gamma(\rho_1) + \gamma(\rho_2) > 0$; moreover, $\rho_1 > \rho_2$ if and only if $w_1 > w_2$ (or, equivalently, $c_1 < c_2$). Finally, let $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\xi^2/2} d\xi$ denote the CDF of the standard normal distribution and define function¹¹

$$\rho(z) = \gamma^{-1}((\gamma(\rho_1) + \gamma(\rho_2))\Phi(z) - \gamma(\rho_2)). \quad (13)$$

The following lemma provides the result.

Lemma 3. *The following pair of functions $A_{\pm}(z)$ solves (8) and satisfies (9):*

$$\begin{aligned} A_+(z) &= \frac{\gamma(\rho_1) + \gamma(\rho_2)}{2\sqrt{2\pi}} [1 - \rho(z)^2] \rho(z), \\ A_-(z) &= \frac{\gamma(\rho_1) + \gamma(\rho_2)}{2\sqrt{2\pi}} [1 - \rho(z)^2]. \end{aligned} \quad (14)$$

To identify the cases of weak and strong competition, consider the argument of $\gamma^{-1}(\cdot)$ in the definition of function $\rho(z)$ (Equation (13)). The argument is strictly increasing in z and varies between $-\gamma(\rho_2)$ at $z \rightarrow -\infty$ and $\gamma(\rho_1)$ at $z \rightarrow +\infty$. Therefore, when ρ_1 and ρ_2 are both positive, the argument changes sign at a point z_m such that $\Phi(z_m) = \gamma(\rho_2)/(\gamma(\rho_1) + \gamma(\rho_2))$, and hence $A_-(z)$ is unimodal, and there is strong competition with a pivot. When $\rho_1 > 0 \geq \rho_2$, the argument is positive for all z ; therefore, $\rho(z)$ is positive, $A_-(z)$ is decreasing, and there is weak competition with dominant player 1. Finally, when $\rho_2 > 0 \geq \rho_1$, the argument is always negative, $\rho(z)$ is negative, $A_-(z)$ is increasing, and we have weak competition with dominant player 2.

Thus, strong competition arises if and only if the following conditions are satisfied:

$$(1) \quad e^{w_1} + e^{-w_2} > 2, \quad (2) \quad e^{w_2} + e^{-w_1} > 2. \quad (15)$$

Notably, these conditions always hold for $w_1 = w_2$, that is, when the players are symmetric. For asymmetric players, the conditions hold when both w_1 and w_2 are large enough. Thus, strong competition emerges either for players who are close in ability, or, if they are not close, both players' ability is sufficiently large. In particular, when $w_1, w_2 \geq \ln 2 \approx 0.6931$, competition is strong regardless of the degree of asymmetry. When one of Conditions (15) does not hold,¹² weak

competition emerges where the player who has a higher ability is dominant.

The regions of strong and weak competition in the (e^{w_1}, e^{w_2}) space are shown in Figure 1. As the players' abilities decrease, the region of strong competition shrinks around the 45-degree line of equal ability.

Combining the results from this section, we obtain the equilibrium effort.

Proposition 1. *In the Markov perfect equilibrium, the players' efforts in state $(x, t) \in \mathbb{R} \times [0, T)$ are*

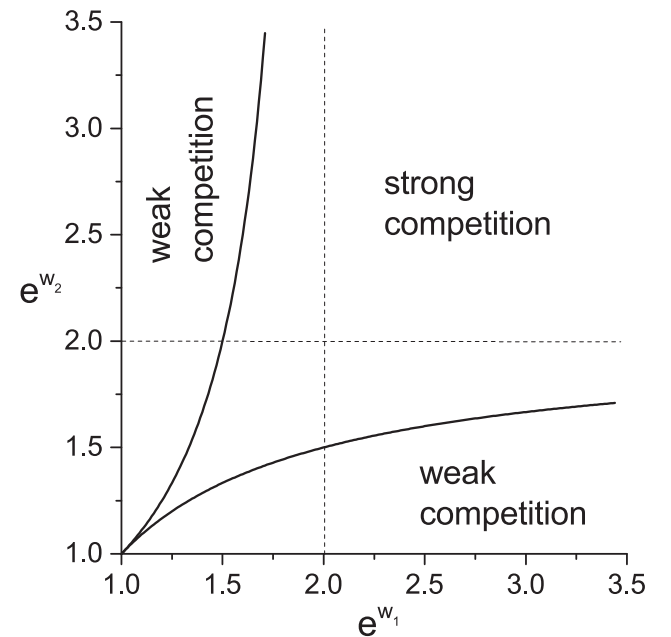
$$\begin{aligned} m_i(x, t) &= \frac{K_i(z)}{\sqrt{2\pi\sigma^2(T-t)}} e^{-z^2/2}, \quad i=1,2, \\ K_{1,2}(z) &= \frac{\sigma^2}{2} [\gamma(\rho_1) + \gamma(\rho_2)] [1 - \rho(z)^2] [1 \pm \rho(z)], \quad z = \frac{x}{\sigma\sqrt{T-t}}. \end{aligned} \quad (16)$$

4. Properties of the Equilibrium

4.1. Effort

Equation (16) has the form $m_i(x, t) = K_i(z) \phi_{N(0, \sigma^2(T-t))}(x)$, where $\phi_{N(0, \sigma^2(T-t))}(\cdot)$ is the PDF of the normal distribution with mean zero and variance $\sigma^2(T-t)$, and $K_i(z)$ are amplitude factors that only depend on the composite variable z . The exponential factor in the normal PDF ensures that effort decays quickly away from $x = 0$. There is a window of width $\sim \sigma\sqrt{T-t}$ around $x = 0$ where effort is nontrivial. This window becomes more narrow, and the peak becomes sharper, as the deadline approaches. Outside of this window of opportunity, both players are essentially cruising toward the deadline, riding the noise. The leader is complacent, and the follower is giving up. However, if the noise brings them inside the window, competition intensifies.

Figure 1. Regions of Ability Parameters for Strong and Weak Competition



The sharp exponential dependence in the normal PDF is primarily responsible for the effects of σ and t on $m_i(x, t)$. Individual effort increases in t (decreases in σ) close to the peak and decreases in t (increases in σ) away from the peak. Aggregate effort, $m_1(x, t) + m_2(x, t)$, behaves similarly. These effects are quite intuitive. Facing more noise, or being further away from the deadline, encourages effort in states away from $x = 0$ because there is a larger chance for the follower to catch up, and discourages effort close to $x = 0$ because it makes sense for the players to wait and see where the noise takes them.

The amplitude factors $K_i(z)$ create a skewness in $m_i(x, t)$. Which of the two players' effort is higher is determined by the sign of $\rho(z)$, that is, from (13), by the sign of $(\gamma(\rho_1) + \gamma(\rho_2))\Phi(z) - \gamma(\rho_2)$. When the players are symmetric, $w_1 = w_2$, the pivot is always at $x_m(t) = 0$, that is, the leader's effort is higher than the follower's effort. To explore the asymmetric case, suppose, for concreteness, that $w_1 > w_2$. Under strong competition, $x_m(t)$ is negative; that is, player 1's dominance zone is larger, and her effort exceeds player 2's effort even in some states where player 1 is trailing. This effect decreases over time as $x_m(t)$ approaches zero. Under weak competition there is no pivot, and player 1's effort is higher than player 2's effort in all states.

These features are illustrated in Figure 2 that shows snapshots in time of the dependence of individual efforts, $m_i(x, t)$, on state variable x . In each row, the left, middle, and right panels show effort at $t = 0$, $t = 0.5$, and $t = 0.9$, respectively, in a contest with total duration $T = 1$. The top row of graphs shows $m_{1,2}(x, t)$ for symmetric players. As expected, these efforts satisfy $m_2(x, t) = m_1(-x, t)$, and the pivot is at $x_m = 0$ at all times. The middle row shows effort for asymmetric players under strong competition. Player 1 has a higher ability, and the pivot is at $x_m(t) < 0$. Finally, the bottom row shows effort under weak competition. Player 1's ability is so much higher that her effort exceeds the effort of player 2 in all states, and there is no pivot.

Aggregate effort, $m_1(x, t) + m_2(x, t)$, is determined by the product $[1 - \rho(z)^2]e^{-z^2/2}$. Under strong competition, the first term is maximized at $x_m(t)$, whereas the second one has a maximum at $x = 0$. However, aggregate effort is unimodal in x at all times (see the proof of Lemma 4 in Section 4.2). The peak of aggregate effort is at a negative (positive) x if player 1 (2) has a higher ability and is located between $x_m(t)$ and zero if a pivot exists.

The dependence on parameters V and c_i is contained entirely in the amplitudes $K_i(z)$. The prize value and cost parameters always enter as ratios, V/c_i , through the composite ability parameters $w_i = V/c_i\sigma^2$. Intuitively, one could expect effort $m_i(x, t)$ to be

increasing in V and decreasing in c_i ; however, the effects turn out to be more complex.

For concreteness, consider $K_1(z) = \frac{\sigma^2 a_0}{2} [1 - \rho(z)^2] [1 + \rho(z)]$. As we show in Appendix A, coefficient $a_0 = \gamma(\rho_1) + \gamma(\rho_2)$ can be expressed through parameters $w_{1,2}$ as

$$a_0 = \frac{w_1 + w_2}{2} + \frac{(e^{w_1+w_2} - 1)(\sinh^2 \frac{w_1}{2} + \sinh^2 \frac{w_2}{2})}{(e^{w_1} - 1)(e^{w_2} - 1)}. \quad (17)$$

For a fixed w_2 , a_0 is U-shaped in w_1 —decreasing for w_1 sufficiently small, and then increasing. Function $\rho(z)$ is increasing in ρ_1 and decreasing in ρ_2 for any z ; hence, it is always increasing in w_1 and decreasing in w_2 . Therefore, $m_i(x, t)$ can be nonmonotone in w_i and w_j , $j \neq i$, that is, it can be nonmonotone in the player's own cost parameter c_i and the other player's cost parameter, c_j . The effect of V is even more complex because changes in V change the effective ability parameters (w_1, w_2) of both players. A simpler setting where the effect of V can be explored is a symmetric contest.

Consider symmetric contests in more detail. Suppose $w_1 = w_2 = w$ and let $m(x, t) = m_1(x, t)$ denote the corresponding equilibrium effort of player 1. Player 2's effort can then be written as $m_2(x, t) = m(-x, t)$. Competition is always strong, with $z_m = 0$, and $\rho_1 = \rho_2 = \tanh \frac{w}{2}$. In this case, (17) gives $a_0 = w + \sinh w$, and Equation (16) then gives

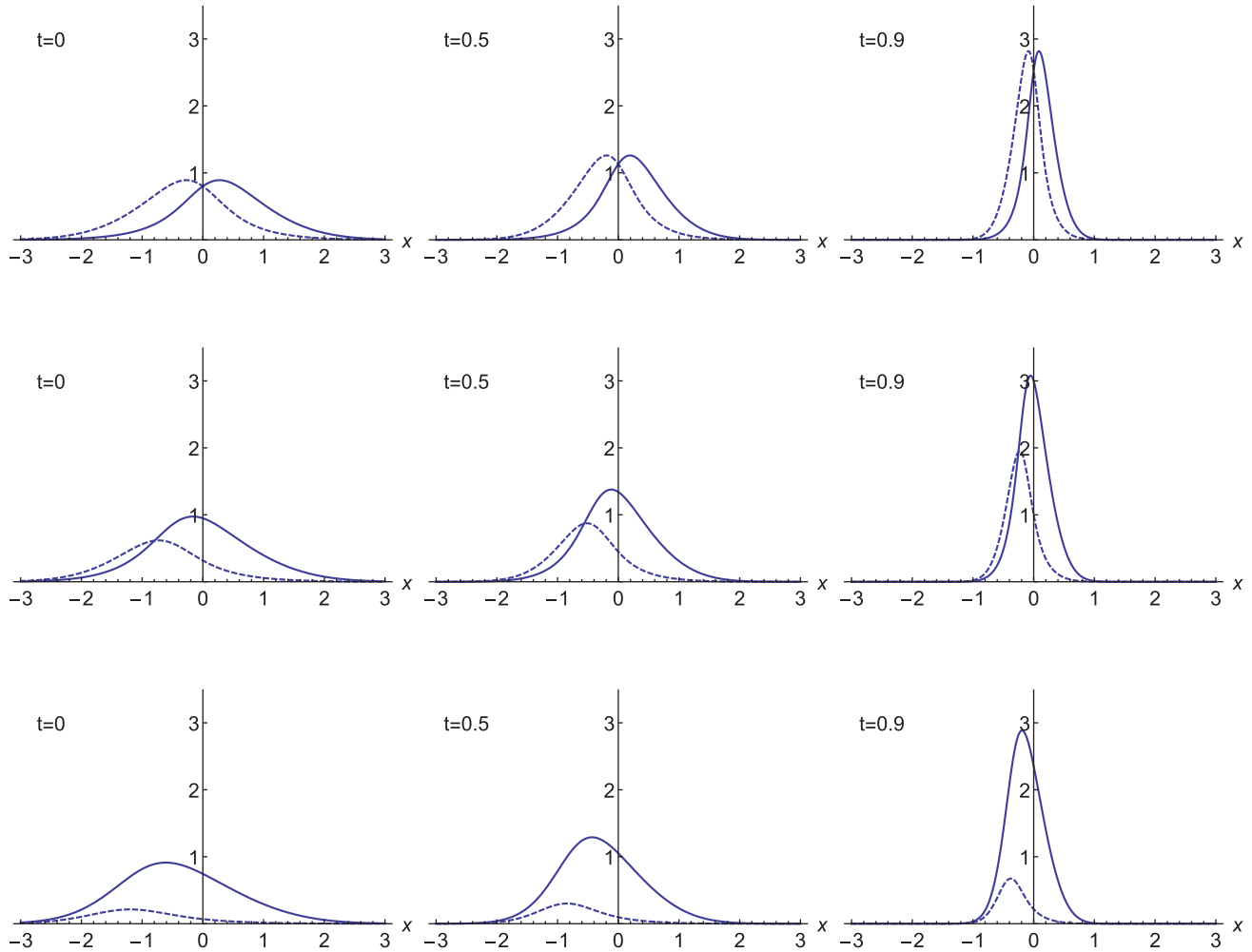
$$m(x, t) = \frac{\sigma^2(w + \sinh w)}{2\sqrt{2\pi\sigma^2(T-t)}} [1 - \rho_s(z)^2] [1 + \rho_s(z)] e^{-z^2/2},$$

$$\rho_s(z) = \gamma^{-1} \left((w + \sinh w) \left(\Phi(z) - \frac{1}{2} \right) \right). \quad (18)$$

It is easy to see that $m(x, t) > m(-x, t)$ for all $x > 0$. Thus, the leader's effort is always higher than the follower's and drops off at higher distances. The maximum of individual effort is reached at a positive lead and not at $x = 0$ as in static contests with additive noise. These features constitute a dynamic momentum effect. At the same time, aggregate effort is maximized at $x = 0$; in other words, the gain in the leader's effort is always lower than the loss in the follower's effort when $x \neq 0$.

To assess how effort depends on w , note that, from the definition of $\rho_s(z)$ in (18), $\gamma(\rho_s(z)) = (w + \sinh w) \left(\Phi(z) - \frac{1}{2} \right)$. Therefore, qualitatively, the dependence of $m(x, t)$ on w can be analyzed by looking at the behavior of function $S(\rho) = \gamma(\rho)(1 - \rho^2)(1 + \rho)$. Indeed, the sign of $S'(\rho)$ is the same as the sign of $\frac{\partial m(x, t)}{\partial w}$. Similarly, the dependence of aggregate effort, $m(x, t) + m(-x, t)$, on w qualitatively follows the shape of function $2\gamma(\rho)(1 - \rho^2)$. The two functions are increasing when ρ is close to zero but decreasing for ρ close enough to ± 1 . A ρ close to these boundaries corresponds to a large enough w as well as a sufficiently

Figure 2. (Color online) Equilibrium Effort of Player 1 (Solid) and Player 2 (Dashed) as a Function of State x , for $t = 0$ (Left), $t = 0.5$ (Middle), and $t = 0.9$ (Right)



Notes. Common parameters: $T = 1$, $V = 1$, $\sigma = 1$. Top row: Symmetric players, $w_1 = w_2 = \ln 5$. Middle row: Asymmetric players, strong competition, $w_1 = \ln 7$, $w_2 = \ln 3$. Bottom row: Asymmetric players, weak competition, $w_1 = \ln 8.5$, $w_2 = \ln 1.5$.

large $|z|$. The latter case corresponds to x further away from zero or t close to the deadline. Intuitively, effort will decrease for large w because competition is too intense. For individual effort, the effect is more pronounced for the follower (when $\rho < 0$).

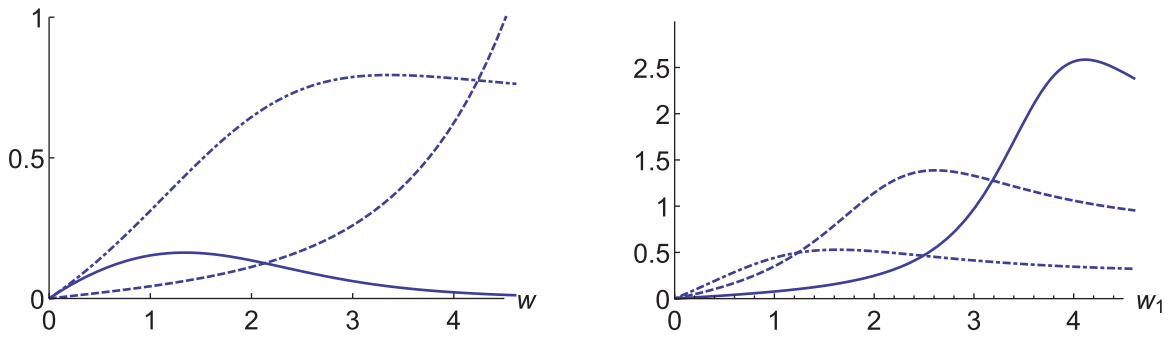
When considering the dependence of effort on parameters and on composite variable $z = x/\sigma\sqrt{T-t}$, many of the properties we identify in the symmetric case are parallel to those in the tug-of-war contest studied by Moscarini and Smith (2011). In particular, the properties that hold in both models are (i) effort is determined by an effective ability variable $w = V/c\sigma^2$; (ii) the follower's effort decreases in z ; (iii) the leader's effort is higher than the follower's effort; and (iv) aggregate effort is single-peaked and maximized at $z = 0$.

Figure 3 shows the dependence of player 1's effort, $m_1(x, t)$, on the ability parameters at different values of x and time $t = 0$.¹³ The left panel shows player 1's

effort in a symmetric contest as a function of common ability w . Effort initially increases in w in all cases, but is monotonically increasing only at $x = 0$. For both $x = -1$ and $x = 1$, the dependence is single-peaked. The turning point where effort starts decreasing in w shifts to the right with x . In this case, an increase in w raises the ability of both players; therefore, competition becomes more intense, and it is harder to catch up with the leader. At $x = -1$, player 1 is behind, and effort starts dropping at a relatively low w . At $x = 1$, player 1 is ahead, and, because of the momentum effect, her effort drops off at a larger distance.

The right panel illustrates the case of asymmetric players. Here, we fix player 2's ability and plot the dependence of $m_1(x, 0)$ on w_1 . The dependence is single peaked for all values of x , including $x = 0$ (indeed, in this case the point $x = 0$ is not special because players are heterogeneous). The turning point at which effort starts decreasing in w_1 is decreasing in x . Intuitively,

Figure 3. (Color online) Equilibrium Effort of Player 1 in Symmetric (Left) and Asymmetric (Right) Contests



Notes. (Left) Equilibrium effort of player 1 as a function of common ability, w , for symmetric players at $x = -1$ (solid), $x = 0$ (dashed, divided by 10), and $x = 1$ (dot-dashed). (Right) Equilibrium effort of player 1 as a function of her ability, w_1 , for asymmetric players with $w_2 = \ln 5$, at $x = -1$ (solid), $x = 0$ (dashed), and $x = 1$ (dot-dashed). Common parameters: $T = 1$, $V = 1$, $\sigma = 1$, $t = 0$.

effort eventually decreases in ability because, when it becomes high, the other player cannot credibly commit to exerting effort at future times. In this case, because the other player's ability is fixed, the turning point shifts to the left with x . Indeed, the larger x is the less likely the other player can win and hence her future effort goes down.

4.2. Total Expected Effort

Given the individual efforts $m_i(x, t)$, let $M(x, t)$ denote total expected effort in the contest going forward from the current state $X_t = x$:

$$M(x, t) = \mathbb{E} \left[\int_t^T [m_1(X_\tau, \tau) + m_2(X_\tau, \tau)] d\tau \mid X_t = x \right].$$

From (1), the HJB equation for $M(x, t)$ is

$$M_t + \frac{\sigma^2}{2} M_{xx} + M_x [m_1(x, t) - m_2(x, t)] + m_1(x, t) + m_2(x, t) = 0. \quad (19)$$

Using (10), we have

$$m_1(x, t) \pm m_2(x, t) = \frac{2\sigma A_{\pm}(z)e^{-z^2/2}}{\sqrt{T-t}};$$

therefore, we look for a solution to Equation (19) in the form $M(x, t) = \sigma q(z)\sqrt{T-t}$. This gives

$$M_t = -\frac{\sigma q(z)}{2\sqrt{T-t}} + \frac{\sigma q'(z)z}{2\sqrt{T-t}}, \quad M_x = q'(z),$$

$$M_{xx} = \frac{q''(z)}{\sigma\sqrt{T-t}},$$

and (19) becomes an ODE for function $q(z)$:

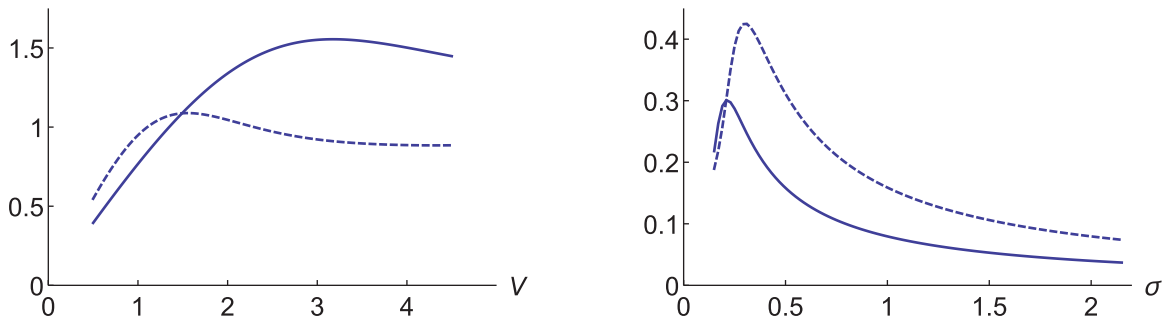
$$q''(z) + \left(4A_+(z)e^{-z^2/2} + z \right) q'(z) - q(z) + 4A_-(z)e^{-z^2/2} = 0. \quad (20)$$

By construction, $q(z)$ must vanish as $z \rightarrow \pm\infty$. The following lemma shows it has the expected properties.

Lemma 4. Suppose a continuous function $q(z)$ solves the boundary value problem (20) with $q(\pm\infty) = 0$. Then, (i) $q(z) > 0$ for all $z \in \mathbb{R}$; and (ii) $q(z)$ unimodal.

Although we are unable to find a closed-form solution to (20), the boundary value problem can be solved numerically.¹⁴ We first assume $c_1 = c_2 = c$ and explore how $M(0, 0) = \sigma\sqrt{T}q(0)$ —total expected effort in a fully symmetric contest—changes with parameters. Examples of the dependence of $M(0, 0)$ on V , for a fixed σ , and on σ , for a fixed V , are shown in Figure 4. As seen from the two graphs, total expected effort is nonmonotone in both parameters. The shape of the curves shows that a principal maximizing $M(0, 0)$ or $M(0, 0) - V$ should choose an optimal value of V that is increasing in σ . If the principal controls σ , she should choose a level of σ increasing in V .¹⁵ Intuitively, although a higher σ dilutes the incentives in states close to $x = 0$, it helps incentivize effort in states further away. When V is large, the contest quickly moves outside the window of opportunity and those more distant states contribute more to total effort.

Examples of total expected effort for contests with asymmetric players are shown in Figure 5. Here, we normalize the duration of the contest to $T = 1$ and show the dependence of $M(x, 0) = \sigma q(z)$ —total expected effort at the start of the contest—on composite variable $z = \frac{x}{\sigma}$. As predicted by Lemma 4, the dependence is single-peaked. In the left panel, we fix player 2's cost at $c_2 = 1$, and show $M(x, 0)$ as player 1's cost decreases from $c_1 = 1$ (symmetric contest) to $c_1 = \frac{1}{2}$ to $c_1 = \frac{1}{3}$. As player 1 becomes stronger, the mode of $\sigma q(z)$ shifts to the left, the peak becomes sharper and more skewed, and the height of the peak increases. At the same time, $M(0, 0)$ —total expected effort for a contest starting at $x = 0$ —is going down. This shows that potentially unbounded total effort can be reached in the contest as one of the players becomes stronger if appropriate asymmetry—a *head-start*—is created at the

Figure 4. (Color online) Total Expected Effort, $M(0, 0)$, in Symmetric Contests

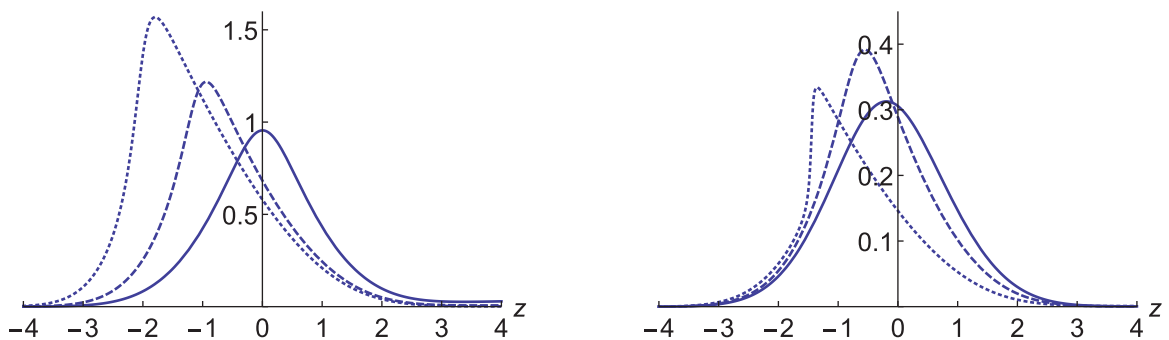
Notes. (Left) $M(0, 0)$ as a function of V for $\sigma = 1$ (solid) and $\sigma = 0.7$ (dashed). (Right) $M(0, 0)$ as a function of σ for $V = 0.1$ (solid) and $V = 0.2$ (dashed). The graphs are obtained by numerically solving boundary value problem (20). The duration of the contest is $T = 1$ and $c = 1$.

beginning. We discuss optimal head-starts and their implications in more detail in Section 5.2.

The right panel in Figure 5 shows $\sigma q(z)$ for a contest with heterogeneous players for different values of σ . The cost parameters are fixed so that player 1 is stronger. As σ decreases, the peak becomes sharper and shifts to the left, similar to the effect of increasing ability. However, the height of the peak is nonmonotone. This effect is similar to the one observed for symmetric contests in the right panel of Figure 4. As noise intensity decreases, the contest becomes more meritocratic. At first, this leads to a higher total effort (under an optimal head-start), but too much meritocracy is detrimental: A lower σ intensifies competition within the window of opportunity, but then the state leaves the window very quickly and rarely comes back.

This result is similar in spirit to the finding of Morgan et al. (2019) that imperfect meritocracy, that is, a less than perfectly discriminatory winner determination process, can be effort-maximizing in static tournaments. The mechanisms behind the two effects are different though. In a symmetric static contest, when

noise becomes too small, competition intensifies to the point where players' equilibrium payoffs are pushed down to zero, and players start dropping out of the contest. There is full dissipation, that is, the total cost of effort of the remaining players is equal to the prize. Therefore, when the cost of effort is strictly convex, the resulting aggregate effort is decreasing as its (fixed) cost is divided among a smaller number of active players. However, when there is heterogeneity, effort may go up because the remaining players have higher abilities than those who drop out. Thus, perfect meritocracy can be optimal in static contests when heterogeneity is in an intermediate range. In contrast, in a dynamic contest, the reduction in aggregate effort for low σ occurs because of the dynamic momentum effect. A large burst of effort at the start of the contest is followed by a period of essentially zero effort as the state is unlikely to ever return inside the window of opportunity when noise is small. Because the cost of effort is strictly convex, such outbursts are very costly compared with when effort is spread out over time; therefore, aggregate effort is reduced.

Figure 5. (Color online) Total Expected Effort at the Start of the Contest, $M(x, 0) = \sigma q(z)$, as a Function of $z = \frac{x}{\sigma}$, for Heterogeneous Players

Notes. The duration of the contest is $T = 1$ and $V = 1$. (Left) $\sigma = 0.7$, and player 2's cost is fixed at $c_2 = 1$; player 1's cost is $c_1 = 1$ (solid), $c_1 = \frac{1}{2}$ (dashed) and $c_1 = \frac{1}{3}$ (dotted). (Right) $c_1 = 5$ and $c_2 = 8$; the noise level is $\sigma = 0.4$ (solid), $\sigma = 0.25$ (dashed) and $\sigma = 0.17$ (dotted). The graphs are obtained by numerically solving boundary value problem (20).

4.3. Probability of Winning and the Role of Early Success

Let $P(x, t) = \mathbb{E}(\mathbb{1}_{X_T > 0} | X_t = x)$ denote the equilibrium probability of player 1 winning the contest starting at state (x, t) . From (1), the HJB equation for its evolution is

$$P_t + \frac{\sigma^2}{2} P_{xx} + P_x[m_1(x, t) - m_2(x, t)] = 0,$$

subject to boundary and terminal conditions $P(-\infty, t) = 0$, $P(\infty, t) = 1$, $P(x, T) = 1$ for $x > 0$, and $P(x, T) = 0$ for $x < 0$.

Similar to Section 4.2, we look for a solution in the form $P(x, t) = p(z)$. The derivatives of $P(x, t)$ then scale as in (5), and the HJB equation becomes an ODE:

$$p''(z) + (4A_+(z)e^{-z^2/2} + z)p'(z) = 0, \quad (21)$$

with boundary conditions $p(-\infty) = 0$ and $p(\infty) = 1$. Remarkably, function $\rho(z)$, Equation (13), satisfies this equation, and the following proposition provides the result.

Proposition 2. *The equilibrium probability of player 1 winning the contest starting at state (x, t) is*

$$P(x, t) = p(z) = \frac{\rho(z) + \rho_2}{\rho_1 + \rho_2}, \quad z = \frac{x}{\sigma\sqrt{T-t}}. \quad (22)$$

When players are symmetric and abilities are low, $p(z) \approx \Phi(z)$. This is because effort is low and the evolution of X_t is essentially free diffusion. As abilities increase, $p(z)$ becomes sharper at the pivot. Figure 6 shows $p(z)$ for symmetric players (left) and asymmetric players (right), for different ability compositions.

The sharp dependence of winning probability on x highlights the role of early success in the contest. When players are symmetric, a lead of order σ is very difficult to overcome even when abilities are very low, more so as time progresses. As abilities increase, the role of early success increases as well, as the window

within which $p(z)$ is not close to zero or one becomes even more narrow. When player 1 has a higher ability, her probability of winning is above $\frac{1}{2}$ at $x = 0$; however, at the pivot, if one exists, $p(z_m) = (\rho_2/(\rho_1 + \rho_2)) < (1/2)$. This implies that player 1's effort is higher than player 2's even in some states, to the right of the pivot, where player 1's probability of winning is lower than player 2's.

4.4. Value Functions

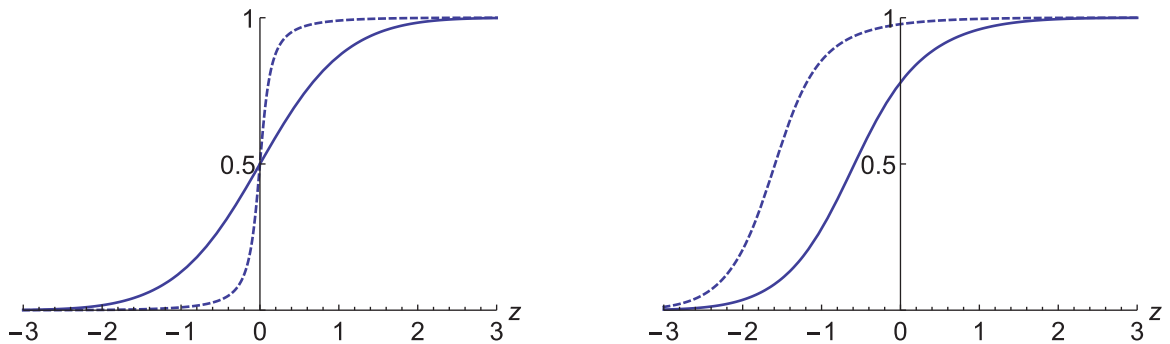
Recall from Section 3 that the equilibrium value functions in state (x, t) are $v_i(x, t) = VF_i(z)$, where functions $F_1(z)$ and $F_2(z)$ are strictly increasing and decreasing, respectively, and, from the boundary conditions at $\pm\infty$, are given by $F_1(z) = \int_{-\infty}^z f_1(\xi) d\xi$ and $F_2(z) = \int_z^{\infty} f_2(\xi) d\xi$. Furthermore, $f_i(z) = A_i(z)e^{-z^2/2}$ and $A_{1,2}(z) = (A_+(z) \pm A_-(z))/w_{1,2}$, where functions $A_{\pm}(z)$ are given by Lemma 3. In the characterization of $v_i(x, t)$, it is convenient to use the equilibrium probabilities of winning, $p_1(z) = p(z)$ and $p_2(z) = 1 - p(z)$, where $p(z)$ is given by Equation (22). The result is as follows.

Proposition 3. *The equilibrium value functions in state (x, t) are*

$$v_i(x, t) = -c_i \sigma^2 \ln[1 - (1 - e^{-w_i})p_i(z)], \quad z = \frac{x}{\sigma\sqrt{T-t}}, \quad i = 1, 2. \quad (23)$$

As expected, player i 's value function is strictly increasing in $p_i(z)$, tending to zero and to V for $p_i(z)$ approaching zero and one, respectively. When w_i is small, we obtain, using the first-order Taylor expansion, $v_i(x, t) \approx c_i \sigma^2 w_i p_i(z) = V p_i(z)$. In general, the value function can be written as $v_i(x, t) = V p_i(z) - C_i(x, t)$, where $C_i(x, t)$ is the expected total cost of future effort.

Figure 6. (Color online) Player 1's Probability of Winning, $p(z)$, as a Function of Composite Variable $z = \frac{x}{\sigma\sqrt{T-t}}$



Notes. (Left) Symmetric contests with $w_1 = w_2 = 1$ (solid) and $w_1 = w_2 = 4$ (dashed). (Right) Asymmetric contests with $w_1 = 2$, $w_2 = 1$ (solid) and $w_1 = 4$, $w_2 = 1$ (dashed).

Thus, player i 's total cost is zero to the first order in w_i .

It is straightforward to see that $v_i(x, t)$ is decreasing in c_i , because player i with a cost $c'_i < c_i$ can achieve a higher value by simply imitating a player with cost c_i . This property can be used to also show that $p_i(z)$, the probability of player i winning the contest, is decreasing in c_i and, because $p_1(z) + p_2(z) = 1$, it implies $p_i(z)$ is increasing in the cost parameter of the other player.

Lemma 5. *Functioning $p_i(z)$ is decreasing in c_i and increasing in c_j , $j \neq i$.*

In a symmetric contest, with $c_1 = c_2 = c$ and $w_1 = w_2 = w$, at any symmetric state $(0, t)$ we have $p_i(0) = \frac{1}{2}$ and $v_i(0, t) = c\sigma^2 \ln(2/(1 + e^{-w}))$, $i = 1, 2$. The players' expected revenue in symmetric states is $V/2$; therefore, their expected costs are $C_i(0, t) = (V/2) - v_i(0, t) = c\sigma^2 \text{Incosh}(w/2)$. The following properties follow directly from the analysis of these equations.

Lemma 6. *The equilibrium value function in the symmetric contest, $v_1(0, t) = v_2(0, t)$, is increasing in V , σ , and c .*

These effects are rather intuitive. The players obtain a fixed expected revenue $\frac{V}{2}$, while their costs adjust to parameters. Although $C_i(0, t)$ is increasing in V , it does so slower than $\frac{V}{2}$. The effect of σ is expected because players have less of an incentive to spend on competition as the contest becomes more noisy. The effect of c is similar: Effort adjusts in such a way that its total cost is decreasing in c .

5. Contest Design

In this section, we study two approaches to contest design that can be undertaken by a principal whose goal is to maximize total expected effort in the contest. One is splitting the contest, and the other is using an optimal head-start when players are heterogeneous.

5.1. Splitting the Contest

For simplicity, in this section we consider fully symmetric contests of identical players, $c_1 = c_2 = c$, starting at $x_0 = 0$. As discussed in Section 4.2, total expected effort is $M(0, 0) = \sigma\sqrt{T}q(0)$, where $q(0)$ is single-peaked in prize V (Figure 4). When V is small, the players have little incentive to exert effort, whereas when V is large, competition at x close to zero is very intense and the state quickly moves away from $x = 0$ where effort remains low.

These properties suggest that the principal may benefit from splitting the contest into multiple subcontests. Suppose we split the contest into $k \geq 2$ identical subcontests, where each subcontest lasts T/k and awards a winner with prize V/k . The resulting total effort is $M^{(k)}(V) = k\sigma\sqrt{T/k}q(0; w/k)$, where, abusing

notation a bit, $q(z; w)$ denotes the value of $q(z)$ for a symmetric contest with effective ability w .

As seen from this simple exercise, splitting the contest can increase total effort in two ways. First, having multiple shorter contests is beneficial because the dependence of total effort on contest duration is concave: $k\sqrt{T/k} > \sqrt{T}$ for $k > 1$. Second, if the prize budget V is large enough, it is possible that $q(0; w) < q(0; w/k)$ for some $k > 1$. Although the first effect can be reproduced by splitting the contest further, the second effect cannot persist indefinitely because, ultimately, each subcontest prize V/k becomes small enough so that $q(0; w/k)$ starts to decrease with k . To understand what happens when k becomes large, we need to explore how $q(0; w)$ scales with w when w is small. The result is given by the following lemma.

Lemma 7. *To the first order in w , total effort in the symmetric contest is*

$$M(x, t) = \frac{2w\sigma\sqrt{T-t}}{\sqrt{2\pi}} e^{-z^2/2} + o(w), \quad z = \frac{x}{\sigma\sqrt{T-t}}. \quad (24)$$

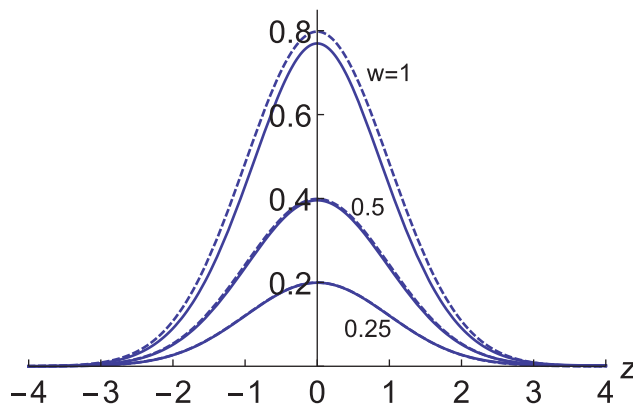
Lemma 7 shows that when V is small, total effort in the contest is linear in V . Therefore, for k large enough, we have

$$M^{(k)}(V) = \sigma\sqrt{Tk}q\left(0; \frac{V}{k}\right) \approx \sigma\sqrt{Tk} \frac{2V}{kc\sigma^2\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi k}} \frac{V\sqrt{T}}{c\sigma},$$

at which point splitting the contest further leads to a reduction in total effort $\propto k^{-1/2}$. Thus, there exists an optimal number of subcontests, k^* , that maximizes $M^{(k)}(V)$. When the total prize budget V is small, $k^* = 1$, but when V is large enough, we have $k^* > 1$. For example, in a contest with $T = 1$, $c = 1$ and $\sigma = 1$ we have $k^* = 1$ for $V = 2$ but $k^* = 2$ for $V = 4$.

Intuitively, splitting the contest up to a point helps for the following reason. As discussed previously, when the contest starts at $x_0 = 0$, effort is high initially but then drops as the state moves away from $x = 0$. Moreover, because the dynamic momentum effect pushes the state away from $x = 0$, this reduction in effort can only be overcome with a large (and unlikely) fluctuation of noise. Splitting the contest allows the principal to artificially *reset* the players back to $x = 0$, where a large burst of effort is guaranteed to happen again. The cost of doing so is, of course, that the prize in these smaller subcontests becomes smaller. Eventually, splitting the contest stops being beneficial because the gain from the reset effect scales as $\sqrt{k}$, whereas the diminishing prize effect scales as $1/k$ for large enough k .¹⁶

Figure 7. (Color online) Total Expected Effort in a Symmetric Contest for Different Values of $w_1 = w_2 = w$



Notes. Solid curves are obtained by numerically solving the boundary value problem (20), and dashed curves are based on Equation (24). Common parameters: $T = 1$, $\sigma = 1$.

It is worth noting that the linear approximation for total effort in Lemma 7 works pretty well for small to moderate values of w . Figure 7 shows the dependence of total effort in the symmetric contest with $T = 1$, $M(x, 0) = \sigma q(z)$, on $z = x/\sigma$, using a high-precision numerical solution of (20) and Equation (24). The agreement for $w = 0.25$ and $w = 0.5$ is excellent; it becomes a bit worse for $w = 1$.

5.2. Optimal Head-Starts

The analysis of total effort for symmetric players in Section 4.2 shows that there is only so much a principal can achieve by manipulating the parameters of the contest. Recall that the symmetric total effort is $M(0, 0) = \sigma \sqrt{T} q(0; w)$. As seen from Figure 4, for a fixed σ , the dependence of $q(0; w)$ on V/c is single-peaked, that is, it is impossible to raise total effort beyond a certain level by increasing the prize or increasing the players' abilities. As discussed earlier, this upper bound arises because of the dynamic momentum effect: When V/c is large, effort is very high initially, but drops off quickly, so that the expectation of its integral over time remains finite.

It is possible to increase $M(0, 0)$ by simultaneously increasing σ and V/c so that w remains constant. This way, the contest becomes more noisy, which helps incentivize players in states away from $x = 0$, and at the same time effort is relatively cheap so that in those states the trailing player still has a decent chance of catching up, which keeps aggregate effort high. From a managerial perspective, this approach is not practical in many settings where luck or the noisiness of performance evaluations are difficult to control. Additionally, increasing V/c is costly because it requires awarding higher prizes or hiring more skilled workers.

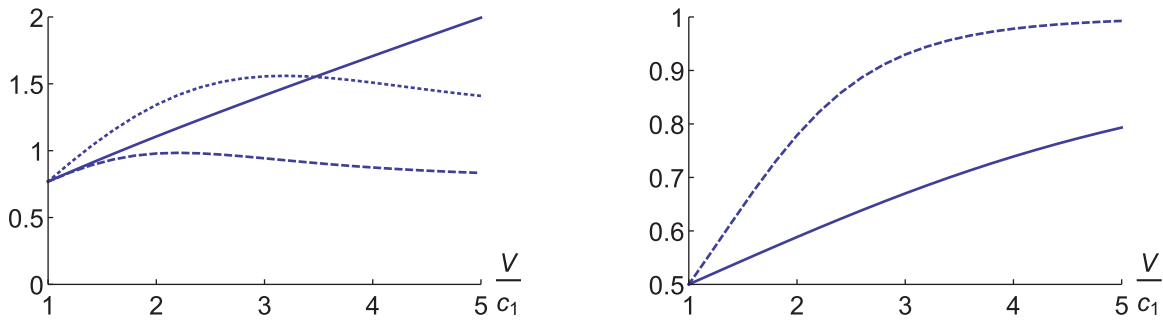
However, as suggested by the analysis of total effort for heterogeneous players—see Figure 5—high total effort in the contest can be generated by appropriately choosing a head-start x_0 , to provide an initial advantage to the weaker player. The optimal (effort-maximizing) head-start is $x_0^* = \sigma z_0 \sqrt{T}$, where z_0 is the unique maximum of $q(z)$. The left panel in Figure 8 shows total effort in contests where V , σ and player 2's cost parameter, c_2 , are fixed so that $w_2 = 1$, whereas player 1's cost parameter, c_1 , is changing so that V/c_1 varies between 1 and 5. For each value of c_1 , we compute the optimal head-start x_0^* and the corresponding total effort $M(x_0^*, 0)$. For comparison, the figure also shows $M(0, 0)$ —total effort without a head-start. As seen from the figure, although $M(0, 0)$ declines for large V/c_1 , $M(x_0^*, 0)$ keeps increasing. The right panel in Figure 8 shows $P(x_0^*, 0)$ and $P(0, 0)$ —player 1's equilibrium probability of winning the contest with the optimal head-start and without a head-start. Although the latter probability rises sharply with $\frac{V}{c_1}$ and quickly approaches one, $P(x_0^*, 0)$ increases much slower and remains well below one even when $c_1/c_2 = 1/5$.

Intuitively, head-starts help increase total effort for two reasons. First, the stronger player has to exert at least some effort to catch up and to move the state past $x = 0$, more so the higher her ability (and hence, the larger the optimal head-start). Second, although the weaker player's probability of winning is below $\frac{1}{2}$ with the optimal head-start, it is still large enough for her to be willing to exert nontrivial effort trying to win. The stronger player then has to respond with a larger effort as well. This intuition also explains the skewness in $q(z)$ (Figure 5): When a head-start is larger than $|x_0^*|$, total effort falls off to zero sharply because the stronger player knows that the weaker player will be able to fend her off and does not even start to compete seriously unless she receives a large positive shock, which is unlikely. In contrast, when a head-start is below $|x_0^*|$, effort falls off much slower because the stronger player still has to surpass $x = 0$ to win.

Interestingly, our analysis shows that it may be better for the principal to have a contest between a low-ability and a high-ability player with an optimal head-start compared with a contest between two high-ability players. The dotted curve in the left panel of Figure 8 shows total effort in the latter case. Here, $c_2 = c_1$ and both players' cost parameters are decreasing. Having more skilled competitors is helpful at first, but when their abilities are too high competition becomes counterproductive.

6. Discussion and Conclusions

Many dynamic contests, from competition in organizations to political campaigns, have (i) a fixed deadline, (ii) participants observing each other's progress

Figure 8. (Color online) Total Expected Effort (Left) and Probability of Player 1 Winning (Right)

Notes. (Left) Total expected effort for heterogeneous players with $\frac{V}{c_2} = 1$ and optimal head-start (solid), for heterogeneous players with $\frac{V}{c_2} = 1$ and without a head-start (dashed), and for symmetric players with $c_2 = c_1$ (dotted). (Right) Probability of player 1 winning for heterogeneous player with $\frac{V}{c_2} = 1$ and optimal head-start (solid), and without a head-start (dashed). Common parameters: $T = 1$, $\sigma = 1$.

in real time, and (iii) performance affected jointly by effort and luck. For such contests, we show that in the Markov perfect equilibrium the players' effort is very sensitive to their relative position, more so the closer the deadline.

When players are symmetric or nearly symmetric ex ante, the key feature of the equilibrium is the dynamic momentum effect. When the score is close to zero, both players' efforts are high and nearly cancel each other; therefore, the direction the contest takes is determined entirely by luck. Once luck resolves the tie, the leader's effort becomes higher than the follower's, pushing the score away from zero and further enhancing the initial effect of luck. Eventually, as the score becomes large, both players' efforts collapse and are unlikely to ever recover.

The important managerial implication is that it may be impossible for a principal to extract higher effort from the agents simply by increasing incentives. When the prize is relatively low, competition is not too intense and a larger prize raises effort throughout the contest. However, increasing the prize beyond a certain level is counterproductive: The initial burst of effort decays quickly, and total effort goes down. Increasing both players' skills has a similar impact. These effects are purely dynamic. Intuitively, in a highly competitive setting, once a player is even a little behind, she has no incentive to exert effort because effort is cheap, and it is easy for the leader to stay ahead; the leader then responds by reducing effort as well. In contrast, in static contests effort can be increased indefinitely by raising the prize or reducing the players' costs.

One approach to increasing effort in the dynamic setting is to increase incentives and simultaneously make the contest more noisy so that "effective abilities"—a combination of prize, costs and noise—remain constant. This result offers an interesting perspective on the debate about excessive risk-taking in highly competitive environments, such as the

financial sector. Typically, the observed correlation between incentives and risk is attributed to the fact that in highly volatile environments agents would have no incentive to exert effort unless they are substantially compensated for success.¹⁷ This is certainly true in our setting as well: When σ is high, effort will be low unless the prize is large enough. Our results, however, also imply that *both* high rewards and large noise are needed to generate high effort in a dynamic setting. If compensation is high but there is not enough volatility in the environment, high effort will not be dynamically sustained.

Restricting information about the state can help mitigate these effects, but it is not always feasible. Even if the principal implements an optimal (non)disclosure policy, the agent who believes she is ahead will have an incentive to disclose her success, to discourage competition. In environments where such leaks cannot be prevented, a simple alternative approach we propose is to split the contest into multiple subcontests with smaller prizes. We show that such subcontests can generate a larger total effort than a single, longer contest for a large prize. The efficiency gain comes from resetting the state to zero, at which point a large burst of effort is guaranteed.

We also show that high total effort can be generated in contests of heterogeneous players with an appropriately chosen head-start for the weaker player. The introduction of such a head-start has two effects. First, the stronger player must exert effort to overcome the deficit. Second, by carefully choosing the head-start, it is possible to raise the weaker player's probability of winning enough to keep her interested as well. As a result, a larger total effort can be reached in a dynamic contest between a weak and a strong player than in a contest between two strong players.

The dynamic momentum effect (also referred to as the dynamic discouragement effect) that we identify is consistent with other models of dynamic contests (Harris and Vickers 1987, Klumpp and Polborn 2006,

Konrad and Kovenock 2009, Cao 2014). There is also empirical evidence, both experimental (Gill and Prowse 2012) and using data from professional sports (Apesteguia and Palacios-Huerta 2010, Iqbal and Krumer 2019). Notably, some studies find a momentum effect in the opposite direction. For example, Genakos and Pagliero (2012) and Berger and Pope (2011), using data from weightlifting and basketball, respectively, find that athletes perform better when they are trailing. These conflicting findings indicate that there may be factors beyond monetary incentives at play, such as choking under pressure (Ariely et al. 2009).

The model we use is rather stylized. Obvious extensions are to consider more general cost functions and discounting, as in Cao (2014), and go beyond $n = 2$ players, for example, as in Seel and Strack (2013). Unfortunately, any of these extensions make the problem intractable. General results on the existence and regularity of solutions to nonlinear parabolic PDEs (Kiselev et al. 2017) can be used to establish the existence of equilibrium in a more general setting with $n = 2$. For $n > 2$ and quadratic costs, the approach used in this paper can be used to characterize the equilibrium. Making further steps in these directions appears rather challenging, however.

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Appendix A. Proofs and Derivations

A.1. Deriving the HJB Equation

Suppose $X_t = x$ and consider a future moment $t + dt$. The value function (2) can be written recursively as

$$v_1(x, t) = \max_{\mu_{1t}} \left[-\frac{c_1 \mu_{1t}^2}{2} dt + \mathbb{E}(v_1(x + dX_t, t + dt)) \right] + o(dt). \quad (\text{A.1})$$

Here, the expectation is over all paths $X_t \rightarrow X_t + dX_t$ starting at $X_t = x$. Using Taylor expansion and retaining terms linear in dt ,

$$\begin{aligned} \mathbb{E}(v_1(x + dX_t, t + dt)) &= v_1(x, t) + v_{1x}(x, t) \mathbb{E}(dX_t) \\ &\quad + \frac{1}{2} v_{1xx}(x, t) \mathbb{E}(dX_t^2) + v_{1t}(x, t) dt + o(dt), \\ &= v_1(x, t) + v_{1x}(x, t) (\mu_{1t} - \mu_{2t}) dt + \frac{\sigma^2}{2} v_{1xx}(x, t) dt + v_{1t}(x, t) dt \\ &\quad + o(dt). \end{aligned}$$

Here, we used the properties of the standard Wiener process: $\mathbb{E}(dW_t) = 0$ and $\mathbb{E}(dW_t^2) = dt$. Plugging this into (A.1), dividing by dt , and considering the limit $dt \rightarrow 0$, obtain (3).

Proof of Lemma 1. For part (i), note that if $f_i(\cdot)$ are continuous then they are C^1 because $f'_i(\cdot)$ are continuous. By

differentiating repeatedly, we conclude that $f_i(\cdot)$ are C^∞ . If $f_1(z_0) = 0$ for some z_0 , then $f'_1(z_0) = 0$. Moreover, by differentiating repeatedly, this also implies $f_1^{(k)}(z_0) = 0$ for all $k \geq 0$. Therefore, $f_1(z)$ must be identically zero, which contradicts the normalization condition. We conclude that $f_1(\cdot)$ does not change sign. The normalization condition then implies $f_1(z) > 0$ for all $z \in \mathbb{R}$. The result for $f_2(z)$ follows similarly.

For part (ii), suppose $f'_1(z_0) = 0$ for some z_0 . The first of Equations (6) then gives $z_0 = -w_1 f_1(z_0) - 2w_2(z_0)$. Differentiating the first of Equations (6) at z_0 and omitting the argument for brevity, obtain

$$\begin{aligned} f''_1(z_0) &= -2w_1 f_1 f'_1 - 2w_2 f'_1 f'_2 - 2w_2 f_1 f'_2 - f'_1 z_0 - f_1, \\ &= -f_1(2w_2 f'_2 + 1) = -f_1[2w_2(-w_2 f_2^2 - 2w_1 f_1 f_2 - f_2 z_0) + 1], \\ &= -f_1[-2w_2^2 f_2^2 - 4w_1 w_2 f_1 f_2 - 2w_2 f_2(-w_1 f_1 - 2w_2) + 1], \\ &= -f_1(2w_2^2 f_2^2 - 2w_1 w_2 f_1 f_2 + 1) < 0. \end{aligned}$$

Here, we used part (i) and the previous expression for z_0 . Thus, any stationary point of $f_1(\cdot)$ is a local maximum. Because of the normalization, $f_1(\cdot)$ must vanish at $\pm\infty$, and hence, it is unimodal. The result for $f_2(\cdot)$ follows similarly. $\square$

A.2. Deriving (8)

Using the relationships $A_\pm(z) = \frac{1}{2}[w_1 A_1(z) \pm w_2 A_2(z)]$ and $A_{1,2}(z) = \frac{1}{w_{1,2}}[A_+(z) \pm A_-(z)]$, and (7), we have

$$\begin{aligned} A'_+(z) &= \frac{1}{2}(w_1 A'_1 + w_2 A'_2) \\ &= -\frac{1}{2}[w_1(w_1 A_1^2 + 2w_2 A_1 A_2) \\ &\quad + w_2(w_2 A_2^2 + 2w_1 A_1 A_2)]e^{-z^2/2}, \\ &= -\frac{1}{2}[A_+^2 + 2A_+ A_- + A_-^2 + 2(A_+^2 - A_-^2) + A_+^2 - 2A_+ A_- + A_-^2 \\ &\quad + 2(A_+^2 - A_-^2)]e^{-z^2/2}, \\ &= (A_-^2 - 3A_+^2)e^{-z^2/2}, \end{aligned}$$

and

$$\begin{aligned} A'_-(z) &= \frac{1}{2}(w_1 A'_1 - w_2 A'_2) \\ &= -\frac{1}{2}[w_1(w_1 A_1^2 + 2w_2 A_1 A_2) \\ &\quad - w_2(w_2 A_2^2 + 2w_1 A_1 A_2)]e^{-z^2/2}, \\ &= -\frac{1}{2}[A_+^2 + 2A_+ A_- + A_-^2 + 2(A_+^2 - A_-^2) - (A_+^2 - 2A_+ A_- + A_-^2) \\ &\quad - 2(A_+^2 - A_-^2)]e^{-z^2/2}, \\ &= -2A_+ A_- e^{-z^2/2}. \end{aligned}$$

Proof of Lemma 2. For part (i), note that $A_-(z)$ is C^∞ and suppose $A_-(z_0) = 0$ at some z_0 . Then, repeatedly differentiating the second equation in (8), we obtain that $A_-^{(k)}(z_0) = 0$ for any $k \geq 0$, which implies $A_-(z) = 0$ identically—a contradiction to the normalization condition. Thus, $A_-(z)$ is sign definite, and the normalization implies it is positive.

For part (ii), suppose $A'_-(z_0) = 0$. This implies $A_+(z_0) = 0$ and $A'_-(z_0) = -2A'_+(z_0)A_-(z_0)e^{-z_0^2/2} = -A_-(z_0)^3 e^{-z_0^2/2} < 0$. Therefore, any stationary point of $A_-(z)$ is a strict local

maximum and hence $A_-(z)$ is either strictly monotone or unimodal.

For part (iii), note that if $A_-(z)$ is strictly monotone then $A'_-(z)$ is sign definite and hence so is $A_+(z)$. If $A_-(z)$ is unimodal with mode z_m , then $A_+(z_m) = 0$, and there are no other points where $A_+(z) = 0$. $\square$

Proof of Lemma 3. We consider separately the two cases identified in part (ii) of Lemma 2 (see the discussion in the text after the lemma).

(i) **Strong competition.** Suppose $A_-(z)$ is unimodal, with a unique mode z_m , and let $a_m = A_-(z_m)$. The constants z_m and a_m will be determined by the normalization conditions. We will now study the Cauchy problem (8) with initial conditions $A_+(z_m) = 0$ and $A_-(z_m) = a_m$.

Consider the domain $z > z_m$ where $A_-(z)$ is strictly decreasing. Then a change of variable $z \mapsto s = A_-(z)$ will produce function $\tilde{A}_+(s) = A_+(A_-^{-1}(s))$. Using (8), its derivative is

$$\tilde{A}'_+(s) = \frac{A'_+(A_-^{-1}(s))}{A'_-(A_-^{-1}(s))} = \frac{s^2 - 3A_+(A_-^{-1}(s))^2}{-2sA_+(A_-^{-1}(s))} = \frac{3\tilde{A}_+^2 - s^2}{2s\tilde{A}_+}. \quad (\text{A.2})$$

This is a homogeneous ODE that can be solved by introducing function $k(s)$ such that $\tilde{A}_+(s) = k(s)s$. This gives

$$k'(s)s + k(s) = \frac{3k(s)^2 - 1}{2k(s)},$$

producing a separable ODE

$$k'(s)s = \frac{k(s)^2 - 1}{2k(s)}.$$

Integration yields $|s| = C|k(s)^2 - 1|$, where C is an arbitrary constant. We know that $A_-(z) > A_+(z)$ for all $z \geq z_m$; otherwise, $A_2(z) = \frac{1}{w_2}[A_+(z) - A_-(z)]$ would be nonnegative for some z , contradicting part (i) of Lemma 1. Moreover, because $A_-(z) > 0$ for all z , we have $s > 0$ and $k(s) < 1$. From the initial condition $\tilde{A}_+(a_m) = 0$ we have $C = a_m$, producing $s = a_m(1 - k(s)^2)$.

There are two solutions, $k(s) = \pm \sqrt{1 - \frac{s}{a_m}}$, corresponding to two branches of $\tilde{A}_+(s)$ opposite in sign. For $z \geq z_m$, the positive branch yields the solution satisfying $A_+(z) \geq 0$. Thus, we have

$$\tilde{A}_+(s) = s\sqrt{1 - \frac{s}{a_m}}, \quad z \geq z_m. \quad (\text{A.3})$$

We can similarly consider the domain $z \leq z_m$. Note that $A_+(z)$ changes sign exactly once (at z_m), and hence to the left of z_m it is negative, producing $\tilde{A}_+(s) = -s\sqrt{1 - \frac{s}{a_m}}$. Combining the two branches, we obtain the relationship between $A_+(\cdot)$ and $A_-(\cdot)$:

$$A_+(z) = \text{sgn}(z - z_m)A_-(z)\sqrt{1 - \frac{A_-(z)}{a_m}}. \quad (\text{A.4})$$

Plugging (A.4) into the second equation in (8), we obtain a separable ODE for $A_-(z)$:

$$A'_-(z) = -2\text{sgn}(z - z_m)e^{-z^2/2}A_-(z)^2\sqrt{1 - \frac{A_-(z)}{a_m}}. \quad (\text{A.5})$$

Consider $z \geq z_m$. With initial condition $A_-(z_m) = a_m$, Equation (A.5) can be integrated as

$$\int_{A_-}^{a_m} \frac{ds}{s^2\sqrt{1 - \frac{s}{a_m}}} = 2 \int_{z_m}^z e^{-\xi^2/2} d\xi = 2\sqrt{2\pi}[\Phi(z) - \Phi(z_m)]. \quad (\text{A.6})$$

The integral on the left-hand side can be written as $\frac{1}{a_m}I_2\left(\frac{A_-}{a_m}\right)$, where

$$\begin{aligned} I_2(a) &= \int_a^1 \frac{ds}{s^2\sqrt{1-s}} = 2 \int_0^{\sqrt{1-a}} \frac{dy}{(1-y^2)^2} \\ &= 2 \int_0^{\sqrt{1-a}} \frac{dy}{(1-y)^2(1+y)^2} \\ &= \frac{1}{2} \int_0^{\sqrt{1-a}} \left[\frac{1}{(1-y)^2} + \frac{1}{(1+y)^2} + \frac{1}{1-y} + \frac{1}{1+y} \right] dy, \\ &= \frac{1}{2} \left(\frac{1}{1-y} - \frac{1}{1+y} + \ln \frac{1+y}{1-y} \right) \Big|_0^{\sqrt{1-a}} = \frac{\sqrt{1-a}}{a} + \frac{1}{2} \ln \frac{1+\sqrt{1-a}}{1-\sqrt{1-a}} \\ &= \gamma(\sqrt{1-a}), \end{aligned}$$

with $\gamma(u)$ given by (11). Thus, Equation (A.6) becomes $\frac{1}{a_m}\gamma\left(\sqrt{1 - \frac{A_-}{a_m}}\right) = 2\sqrt{2\pi}[\Phi(z) - \Phi(z_m)]$, which produces

$$A_-(z) = a_m[1 - \gamma^{-1}(2\sqrt{2\pi}a_m(\Phi(z) - \Phi(z_m)))^2]. \quad (\text{A.7})$$

For $z \leq z_m$, the integral of Equation (A.5) is the same as (A.6), except with a minus sign on the right-hand side, which becomes $\frac{1}{a_m}\gamma\left(\sqrt{1 - \frac{A_-}{a_m}}\right) = 2\sqrt{2\pi}[\Phi(z_m) - \Phi(z)]$, and hence $A_-(z) = a_m[1 - \gamma^{-1}(2\sqrt{2\pi}a_m(\Phi(z_m) - \Phi(z)))^2]$. However, because $\gamma^{-1}(\cdot)$ is an odd function, the result is still (A.7). That is, $A_-(z)$ is given by (A.7) for all z .

To find z_m and a_m , we use the normalization conditions (9). From (A.7), we have $A_-(\pm\infty) = a_m(1 - \rho_{1,2}^2)$, where we define $\rho_1 = \gamma^{-1}(2\sqrt{2\pi}a_m(1 - \Phi(z_m)))$ and $\rho_2 = \gamma^{-1}(2\sqrt{2\pi}a_m\Phi(z_m))$. From (A.5),

$$\begin{aligned} \int_{-\infty}^{+\infty} A_-(z)e^{-z^2/2} dz &= \int_{-\infty}^{z_m} A_-(z)e^{-z^2/2} dz + \int_{z_m}^{+\infty} A_-(z)e^{-z^2/2} dz, \\ &= \int_{A_-(-\infty)}^{a_m} \frac{ds}{2s\sqrt{1 - \frac{s}{a_m}}} - \int_{a_m}^{A_-(+\infty)} \frac{ds}{2s\sqrt{1 - \frac{s}{a_m}}} \\ &= \frac{1}{2}I_1\left(\frac{A_-(-\infty)}{a_m}\right) + \frac{1}{2}I_1\left(\frac{A_-(+\infty)}{a_m}\right), \end{aligned}$$

where

$$I_1(a) = \int_a^1 \frac{ds}{s\sqrt{1-s}} = 2 \int_0^{\sqrt{1-a}} \frac{dy}{1-y^2} = \ln \frac{1+\sqrt{1-a}}{1-\sqrt{1-a}}.$$

Equation (9) then gives

$$\ln \frac{1 + \sqrt{1 - \frac{A_-(-\infty)}{a_m}}}{1 - \sqrt{1 - \frac{A_-(-\infty)}{a_m}}} + \ln \frac{1 + \sqrt{1 - \frac{A_-(+\infty)}{a_m}}}{1 - \sqrt{1 - \frac{A_-(+\infty)}{a_m}}} = w_1 + w_2,$$

or

$$\ln \frac{1+\rho_2}{1-\rho_2} + \ln \frac{1+\rho_1}{1-\rho_1} = w_1 + w_2.$$

For $A_+(z)$ we have, from (A.4), (A.5), and (9),

$$\begin{aligned} \int_{-\infty}^{+\infty} A_+(z) e^{-z^2/2} dz &= \int_{-\infty}^{z_m} A_+(z) e^{-z^2/2} dz + \int_{z_m}^{+\infty} A_+(z) e^{-z^2/2} dz, \\ &= - \int_{A_-(\infty)}^{a_m} \frac{ds}{2s} - \int_{a_m}^{A_-(+\infty)} \frac{ds}{2s} = \frac{1}{2} \ln \frac{A_-(\infty)}{A_-(+\infty)} = \frac{1}{2} \ln \frac{1-\rho_2^2}{1-\rho_1^2} \\ &= \frac{w_1 - w_2}{2}. \end{aligned}$$

Thus, $\rho_{1,2}$ satisfy the system of equations

$$\frac{(1+\rho_1)(1+\rho_2)}{(1-\rho_1)(1-\rho_2)} = e^{w_1+w_2}, \quad \frac{1-\rho_2^2}{1-\rho_1^2} = e^{w_1-w_2},$$

By multiplying and dividing the two equations, they can be reduced to the system of linear equations

$$\frac{1+\rho_2}{1-\rho_1} = e^{w_1}, \quad \frac{1+\rho_1}{1-\rho_2} = e^{w_2},$$

which has the solution

$$\rho_1 = \frac{e^{w_1+w_2} - 2e^{w_2} + 1}{e^{w_1+w_2} - 1}, \quad \rho_2 = \frac{e^{w_1+w_2} - 2e^{w_1} + 1}{e^{w_1+w_2} - 1}.$$

It is straightforward to see that $\rho_{1,2}$ so defined coincide with (12) and $\rho_{1,2} < 1$ always holds; however, for a_m and z_m to be well defined, we should also require that $\rho_{1,2} > 0$. This leads to restrictions (15). Inverting the definitions of $\rho_{1,2}$ through $\gamma^{-1}(\cdot)$, we further obtain

$$2\sqrt{2\pi}a_m(1 - \Phi(z_m)) = \gamma(\rho_1), \quad 2\sqrt{2\pi}a_m\Phi(z_m) = \gamma(\rho_2).$$

This gives $\Phi(z_m) = \frac{\gamma(\rho_2)}{\gamma(\rho_1) + \gamma(\rho_2)}$ and $a_m = \frac{\gamma(\rho_1) + \gamma(\rho_2)}{2\sqrt{2\pi}}$. Finally, defining $\rho(z)$ as in (13), we obtain the equations in the lemma.

(ii) **Weak competition.** Suppose, without loss, that $A_-(z)$ is strictly increasing. Then a change of variable $z \rightarrow s = A_-(z)$ will again produce function $\tilde{A}_+(s) = A_+(A_-^{-1}(s))$ satisfying (A.2). Because $A'_-(z) > 0$ by assumption, we have $A_+(z) < 0$, and hence the negative branch of the solution applies, $\tilde{A}_+(s) = -s\sqrt{1-Cs}$, where $C > 0$ is a constant to be determined. The second equation in (8) then gives

$$A'_-(z) = 2e^{-z^2/2} A_-(z)^2 \sqrt{1-CA_-(z)}. \quad (\text{A.8})$$

Integrating (A.8) over $(-\infty, z]$, we obtain

$$\int_{A_-(\infty)}^{A_-} \frac{ds}{s^2 \sqrt{1-Cs}} = 2\sqrt{2\pi}\Phi(z),$$

which gives

$$C[\gamma(\tilde{\rho}_2) - \gamma(\sqrt{1-CA_-(z)})] = 2\sqrt{2\pi}\Phi(z), \quad (\text{A.9})$$

where $\gamma(\cdot)$ is defined in (11), and $\tilde{\rho}_2 = \sqrt{1-CA_-(\infty)}$ is a constant to be determined later. Similarly, define also $\tilde{\rho}_1 = \sqrt{1-CA_-(+\infty)} < \tilde{\rho}_2$.

Next, we use normalization conditions (9) to find the unknown constants. The conditions for $A_-(z)$ and $A_+(z)$ together with (A.8) give, respectively,

$$\begin{aligned} \int_{-\infty}^{+\infty} A_-(z) e^{-z^2/2} dz &= \int_{A_-(\infty)}^{A_-(+\infty)} \frac{ds}{2s\sqrt{1-Cs}} \\ &= \frac{1}{2} \ln \frac{1+\tilde{\rho}_2}{1-\tilde{\rho}_2} - \frac{1}{2} \ln \frac{1+\tilde{\rho}_1}{1-\tilde{\rho}_1} = \frac{w_1 + w_2}{2}, \\ \int_{-\infty}^{+\infty} A_+(z) e^{-z^2/2} dz &= - \int_{A_-(\infty)}^{A_-(+\infty)} \frac{ds}{2s} = \frac{1}{2} \ln \frac{A_-(\infty)}{A_-(+\infty)} = \frac{1}{2} \ln \frac{1-\tilde{\rho}_2^2}{1-\tilde{\rho}_1^2} \\ &= \frac{w_1 - w_2}{2}. \end{aligned}$$

Thus, $\tilde{\rho}_{1,2}$ satisfy the equations

$$\frac{(1+\tilde{\rho}_2)(1-\tilde{\rho}_1)}{(1-\tilde{\rho}_2)(1+\tilde{\rho}_1)} = e^{w_1+w_2}, \quad \frac{1-\tilde{\rho}_2^2}{1-\tilde{\rho}_1^2} = e^{w_1-w_2}.$$

Multiplying and dividing these equations, we obtain the system of linear equations

$$\frac{1+\tilde{\rho}_2}{1+\tilde{\rho}_1} = e^{w_1}, \quad \frac{1-\tilde{\rho}_1}{1-\tilde{\rho}_2} = e^{w_2},$$

whose solution is

$$\tilde{\rho}_1 = \frac{-e^{w_1+w_2} + 2e^{w_2} - 1}{e^{w_1+w_2} - 1}, \quad \tilde{\rho}_2 = \frac{e^{w_1+w_2} - 2e^{w_1} + 1}{e^{w_1+w_2} - 1}.$$

Thus, we notice that $\tilde{\rho}_1 = -\rho_1$ and $\tilde{\rho}_2 = \rho_2$, where $\rho_{1,2}$ are defined in (12). Therefore, to have $\tilde{\rho}_{1,2} \in (0, 1)$, we need to require that condition (15) (1) be violated. Hence, $A_-(z)$ increasing corresponds to $w_1 < w_2$, that is, player 2 is the higher ability player in this weak competition case. Symmetrically, the other weak competition case where $w_1 > w_2$ (player 1 has a higher ability) corresponds to a decreasing $A_-(z)$, $\tilde{\rho}_1 = \rho_1$, $\tilde{\rho}_2 = -\rho_2$, and condition (15) (2) violated.

Evaluating (A.9) at $z \rightarrow +\infty$, we have $C[\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1)] = 2\sqrt{2\pi}$, which gives $C = \frac{2\sqrt{2\pi}}{\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1)}$ and

$$\begin{aligned} \gamma(\tilde{\rho}_2) - \gamma(\sqrt{1-CA_-(z)}) &= (\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1))\Phi(z), \\ A_-(z) &= \frac{\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1)}{2\sqrt{2\pi}} [1 - \gamma^{-1}(\gamma(\tilde{\rho}_2) - (\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1))\Phi(z))^2]. \end{aligned}$$

Finally, $A_+(z) = -A_-(z)\sqrt{1-CA_-(z)}$, which gives

$$\begin{aligned} A_+(z) &= - \frac{\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1)}{2\sqrt{2\pi}} [1 \\ &\quad - \gamma^{-1}(\gamma(\tilde{\rho}_2) - (\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1))\Phi(z))^2] \gamma^{-1}(\gamma(\tilde{\rho}_2) \\ &\quad - (\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1))\Phi(z)), \end{aligned}$$

$$\begin{aligned}
&= \frac{\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1)}{2\sqrt{2\pi}} [1 \\
&\quad - \gamma^{-1}((\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1))\Phi(z) - \gamma(\tilde{\rho}_2)^2)] \gamma^{-1}((\gamma(\tilde{\rho}_2) - \gamma(\tilde{\rho}_1))\Phi(z) \\
&\quad - \gamma(\tilde{\rho}_2)).
\end{aligned}$$

From $\tilde{\rho}_1 = \rho_1$ and $\tilde{\rho}_2 = -\rho_2$, we obtain the result. $\square$

A.3. Deriving (17)

From the definition of $\gamma(\cdot)$, Equation (11), we have

$$a_0 = \gamma(\rho_1) + \gamma(\rho_2) = \frac{\rho_1}{1 - \rho_1^2} + \frac{\rho_2}{1 - \rho_2^2} + \frac{1}{2} \ln \frac{(1 + \rho_1)(1 + \rho_2)}{(1 - \rho_1)(1 - \rho_2)}.$$

Using (12), we have

$$\begin{aligned}
\rho_{1,2} &= \frac{e^{w_1+w_2} - 2e^{w_{2,1}} + 1}{e^{w_1+w_2} - 1}, \quad 1 + \rho_{1,2} = \frac{2e^{w_{2,1}}(e^{w_{1,2}} - 1)}{e^{w_1+w_2} - 1}, \\
1 - \rho_{1,2} &= \frac{2(e^{w_{2,1}} - 1)}{e^{w_1+w_2} - 1},
\end{aligned}$$

which gives

$$\begin{aligned}
a_0 &= \frac{(e^{w_1+w_2} - 2e^{w_2} + 1)(e^{w_1+w_2} - 1)}{4e^{w_2}(e^{w_1} - 1)(e^{w_2} - 1)} \\
&\quad + \frac{(e^{w_1+w_2} - 2e^{w_1} + 1)(e^{w_1+w_2} - 1)}{4e^{w_1}(e^{w_1} - 1)(e^{w_2} - 1)} \\
&\quad + \frac{1}{2} \ln \frac{e^{w_2}(e^{w_1} - 1)}{e^{w_2} - 1} \cdot \frac{e^{w_1}(e^{w_2} - 1)}{e^{w_1} - 1}, \\
&= \frac{w_1 + w_2}{2} \\
&\quad + \frac{(e^{w_1+w_2} - 1)(e^{2w_1+w_2} - 2e^{w_1+w_2} + e^{w_1} + e^{w_1+2w_2} - 2e^{w_1+w_2} + e^{w_2})}{4e^{w_1+w_2}(e^{w_1} - 1)(e^{w_2} - 1)}, \\
&= \frac{w_1 + w_2}{2} + \frac{(e^{w_1+w_2} - 1)(e^{w_1} - 2 + e^{-w_2} + e^{w_2} - 2 + e^{-w_1})}{4(e^{w_1} - 1)(e^{w_2} - 1)}, \\
&= \frac{w_1 + w_2}{2} + \frac{(e^{w_1+w_2} - 1)[(e^{w_1/2} - e^{-w_1/2})^2 + (e^{w_2/2} - e^{-w_2/2})^2]}{4(e^{w_1} - 1)(e^{w_2} - 1)},
\end{aligned}$$

and (17) follows.

Proof of Lemma 4. For part (i), suppose $q(z_1) \leq 0$ for some z_1 . Then, from the boundary conditions $q(\pm\infty) = 0$, $q(z)$ must have a local minimum point z_2 such that $q'(z_2) = 0$, $q''(z_2) \geq 0$ and $q(z_2) \leq 0$, which is incompatible with (20).

For part (ii), we first prove that (instantaneous) aggregate effort, $m_1(x, t) + m_2(x, t)$, is unimodal in x . Recall that $m_1(x, t) + m_2(x, t) = \frac{2\sigma A_-(z)e^{-z^2/2}}{\sqrt{T-t}}$; therefore, using (14), it is sufficient to prove that $U(z) = [1 - \rho(z)^2]e^{-z^2/2}$ is unimodal. Suppose z_0 is a stationary point of $U(z)$, that is, $U'(z_0) = 0$. We will show that it has to be a strict local maximum, that is, $U''(z_0) < 0$.

Inverting (13), $\gamma(\rho(z)) = (\gamma(\rho_1) + \gamma(\rho_2))\Phi(z) - \gamma(\rho_2)$. Differentiating twice, obtain

$$\begin{aligned}
\gamma'(\rho(z))\rho'(z) &= \frac{\gamma(\rho_1) + \gamma(\rho_2)}{\sqrt{2\pi}} e^{-z^2/2}, \\
\gamma''(\rho(z))\rho'(z)^2 + \gamma'(\rho(z))\rho''(z) &= -\frac{\gamma(\rho_1) + \gamma(\rho_2)}{\sqrt{2\pi}} ze^{-z^2/2} \quad (10) \\
&= -z\gamma'(\rho(z))\rho'(z).
\end{aligned}$$

This gives

$$\rho''(z) = -z\rho'(z) - \frac{\gamma''(\rho(z))\rho'(z)^2}{\gamma'(\rho(z))}.$$

Differentiating $U(z)$ twice, obtain

$$\begin{aligned}
U'(z) &= [-2\rho(z)\rho'(z) - (1 - \rho(z)^2)z]e^{-z^2/2}, \\
U''(z) &= [-2\rho'(z)^2 - 2\rho(z)\rho''(z) + 4\rho(z)\rho'(z)z \\
&\quad - (1 - \rho(z)^2)(1 - z^2)]e^{-z^2/2}.
\end{aligned}$$

From the first equation, $U'(z_0) = 0$ implies $z_0 = -\frac{2\rho(z_0)\rho'(z_0)}{1 - \rho(z_0)^2}$.

From the definition of $\gamma(u)$, Equation (11), we have $\gamma'(u) = \frac{2}{(1-u^2)^2}$ and $\gamma''(u) = \frac{8u}{(1-u^2)^3}$, implying $\frac{\gamma''(u)}{\gamma'(u)} = \frac{4u}{1-u^2}$, and

$$\rho''(z) = -z\rho'(z) - \frac{4\rho(z)\rho'(z)^2}{1 - \rho(z)^2}. \quad (\text{A.11})$$

At z_0 such that $U'(z_0) = 0$, we have $\rho''(z_0) = -z_0\rho'(z_0) + 2z_0\rho'(z_0) = z_0\rho'(z_0)$ and

$$\begin{aligned}
U''(z_0) &= [-2\rho'(z_0)^2 + 2\rho(z_0)\rho'(z_0)z_0 - (1 - \rho(z_0)^2)(1 - z_0^2)]e^{-z_0^2/2}, \\
&= [-2\rho'(z_0)^2 - (1 - \rho(z_0)^2)z_0^2 - (1 - \rho(z_0)^2)(1 - z_0^2)]e^{-z_0^2/2}, \\
&= [-2\rho'(z_0)^2 - (1 - \rho(z_0)^2)]e^{-z_0^2/2} < 0.
\end{aligned}$$

Next, we prove that $q(z)$ is unimodal. It follows from the boundary conditions and part (i) that there exists a $\hat{z}$ such that $q'(\hat{z}) = 0$. Equation (20) then implies $q''(\hat{z}) = q(\hat{z}) - 4A_-(\hat{z})e^{-\hat{z}^2/2}$. Suppose $\hat{z} \leq z_0$. If $\hat{z}$ is a local minimum, we have $q''(\hat{z}) \geq 0$, and there has to exist a local maximum $z' < \hat{z}$ such that $q'(z') = 0$, $q''(z') \leq 0$ and $q(z') \geq q(\hat{z})$. This is impossible because $A_-(z')e^{-z'^2/2} < A_-(\hat{z})e^{-\hat{z}^2/2}$. A similar contradiction is obtained if we assume $\hat{z} \geq z_0$, in which case, if $\hat{z}$ is a local minimum, there has to exist a local maximum $z'' \geq \hat{z}$. Thus, any stationary point of $q(z)$ is a local maximum, and hence $q(z)$ is unimodal. $\square$

Proof of Proposition 2. It is sufficient to show that $\rho(z)$ satisfies the homogeneous linear ODE (21). Using Equations (A.10) and (A.11) in the proof of Lemma 4, we have

$$\begin{aligned}
\rho''(z) + (4A_+(z)e^{-z^2/2} + z)\rho'(z) &= -\frac{4\rho(z)\rho'(z)^2}{1 - \rho(z)^2} \\
&\quad + 4A_+(z)e^{-z^2/2}\rho'(z), \\
&= \frac{4\rho(z)\rho'(z)}{1 - \rho(z)^2} \left[\frac{a_0(1 - \rho(z)^2)^2 e^{-z^2/2}}{2\sqrt{2\pi}} - \rho'(z) \right] \\
&= \frac{4\rho(z)\rho'(z)}{1 - \rho(z)^2} \left[\frac{a_0 e^{-z^2/2}}{\sqrt{2\pi}\gamma'(\rho(z))} - \rho'(z) \right] = 0.
\end{aligned}$$

The result follows by imposing the boundary conditions. $\square$

Proof of Proposition 3. For player 1, we have

$$\begin{aligned}
v_1(x, t) &= VF_1(z) = V \int_{-\infty}^z f_1(\xi) d\xi = V \int_{-\infty}^z A_1(\xi) e^{-\xi^2/2} d\xi, \\
&= \frac{V}{w_1} \int_{-\infty}^z [A_+(\xi) + A_-(\xi)] e^{-\xi^2/2} d\xi \\
&= c_1 \sigma^2 \left[\int_{-\infty}^z A_+(\xi) e^{-\xi^2/2} d\xi + \int_{-\infty}^z A_-(\xi) e^{-\xi^2/2} d\xi \right].
\end{aligned}$$

Using (8), we have

$$\begin{aligned} \int_{-\infty}^z A_+(\xi) e^{-\xi^2/2} d\xi &= - \int_{-\infty}^z \frac{A'_-(\xi)}{2A_-(\xi)} d\xi = \frac{1}{2} \ln \frac{A_-(-\infty)}{A_-(z)} \\ &= \frac{1}{2} \ln \frac{1 - \rho_2^2}{1 - \rho(z)^2}. \end{aligned}$$

Similarly, following the same steps as in the proof of Lemma 3, we obtain

$$\int_{-\infty}^z A_-(\xi) e^{-\xi^2/2} d\xi = \frac{1}{2} \ln \frac{1 + \rho_2}{1 - \rho_2} + \frac{1}{2} \ln \frac{1 + \rho(z)}{1 - \rho(z)}.$$

Thus, we have, using (22)

$$\begin{aligned} v_1(x, t) &= \frac{c_1 \sigma^2}{2} \ln \frac{(1 - \rho_2^2)(1 + \rho_2)(1 + \rho(z))}{(1 - \rho(z)^2)(1 - \rho_2)(1 - \rho(z))} \\ &= c_1 \sigma^2 \ln \frac{1 + \rho_2}{1 - \rho(z)}, \\ &= c_1 \sigma^2 \ln \frac{1 + \rho_2}{1 - (\rho_1 + \rho_2)p(z) + \rho_2} = c_1 \sigma^2 \ln \frac{1}{1 - \frac{\rho_1 + \rho_2}{1 + \rho_2} p(z)}. \end{aligned}$$

Finally, from (12),

$$\frac{\rho_1 + \rho_2}{1 + \rho_2} = 1 - \frac{1 - \rho_1}{1 + \rho_2} = 1 - e^{-w_1},$$

which produces the result. For player 2, the result follows by symmetry. $\square$

Proof of Lemma 5. For brevity, we drop the arguments of $p_i(z)$ and $v_i(x, t)$, and let $v_i = \frac{v_i}{c_i \sigma^2}$. From (23), we have $p_i = \frac{1 - e^{-v_i}}{1 - e^{-w_i}}$. Differentiating with respect to c_i , obtain

$$\begin{aligned} p_{ic_i} &= \frac{1}{c_i^2 \sigma^2 (1 - e^{-w_i})^2} [e^{-v_i} (v_{ic_i} c_i - v_i) (1 - e^{-w_i}) + e^{-w_i} V (1 - e^{-v_i})], \\ &= \frac{e^{-v_i} v_{ic_i}}{c_i \sigma^2 (1 - e^{-w_i})} \\ &\quad - \frac{1}{c_i (1 - e^{-w_i})^2} [e^{-v_i} v_i (1 - e^{-w_i}) - e^{-w_i} w_i (1 - e^{-v_i})]. \end{aligned}$$

Because $v_{ic_i} < 0$, it is sufficient to show that $e^{-v_i} v_i (1 - e^{-w_i}) - e^{-w_i} w_i (1 - e^{-v_i}) \geq 0$ or, equivalently, $\frac{v_i}{e^{v_i} - 1} \geq \frac{w_i}{e^{w_i} - 1}$. This inequality holds because $v_i < V$, and hence $v_i < w_i$, in all states, and function $u \mapsto \frac{u}{e^u - 1}$ is decreasing in u . $\square$

Proof of Lemma 7. Consider a symmetric contest with $w_1 = w_2 = w$, and look for a solution to the boundary value problem (20) in the form $q(z) = q_0(z) + q_1(z)w + o(w)$. When $w = 0$, we have $A_+(z) = A_-(z) = 0$, Equation (20) is homogeneous, and hence the only solution with zero values at the boundary is the trivial solution $q_0(z) = 0$.

Recall that $\gamma'(u) = \frac{2}{(1 - u^2)^2}$; therefore, $\gamma'(0) = 2$ and $(\gamma^{-1})'(0) = \frac{1}{2}$. To the first order in w , we have from (18):

$$\begin{aligned} \rho_s(z) &= \gamma^{-1}(0) + (\gamma^{-1})'(0) 2w \left(\Phi(z) - \frac{1}{2} \right) + o(w) \\ &= w \left(\Phi(z) - \frac{1}{2} \right) + o(w). \end{aligned}$$

Furthermore, $a_0 = 2w + o(w)$; therefore,

$$\begin{aligned} A_-(z) &= \frac{a_0}{2\sqrt{2\pi}} (1 - \rho_s(z)^2) e^{-z^2/2} = \frac{w}{\sqrt{2\pi}} e^{-z^2/2} + o(w), \\ A_+(z) &= A_-(z) \rho_s(z) = o(w). \end{aligned}$$

Thus, Equation (20) gives for $q_1(z)$,

$$q_1''(z) + z q_1'(z) - q_1(z) + \frac{4}{\sqrt{2\pi}} e^{-z^2/2} = 0.$$

It is easy to verify that $q_1(z) = \frac{2}{\sqrt{2\pi}} e^{-z^2/2}$ is its solution satisfying the boundary conditions $q_1(\pm\infty) = 0$, which implies $q(z) = \frac{2w}{\sqrt{2\pi}} e^{-z^2/2} + o(w)$. $\square$

Appendix B. Zero-Noise Limit

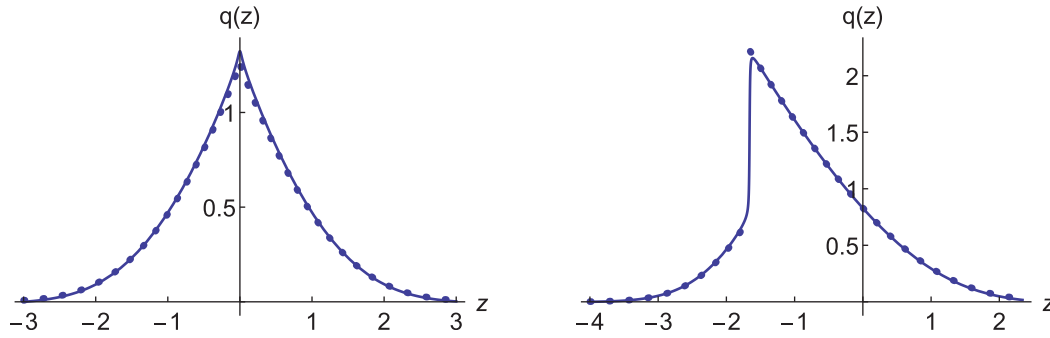
The results in Section 3 suggest that individual and aggregate effort in the contest converge to zero as $\sigma \rightarrow 0$. To verify this conjecture, we consider the contest without noise in more detail. As we show, the result indeed holds for symmetric players. In contrast, when players are heterogeneous, it is possible to extract some effort from the stronger player by introducing a head-start.

Proposition B.1. Suppose $\sigma = 0$ and $c_1 = c_2$. Then there is a subgame-perfect equilibrium in which both players exert no effort. If $x_0 = 0$, each player wins with probability $\frac{1}{2}$, whereas if $x_0 > (<) 0$, player 1 (2) wins with certainty.

Consider first the case when $x_0 = 0$. Then the following strategy, employed by each player, constitutes a sub-game perfect equilibrium: Start with zero effort; match the other player's effort at any time going forward as long as the total cost of effort is below $\frac{V}{2}$. To see why this is an equilibrium, if both players continue to exert zero effort, each player earns $\frac{V}{2}$ in expectation. A profitable deviation should, therefore, lead to a payoff strictly above $\frac{V}{2}$, that is, it has to involve a total cost of effort strictly below $\frac{V}{2}$. However, the other player will be willing to match such effort; therefore, the state will remain at $x = 0$ and the deviating player's payoff will be strictly below $\frac{V}{2}$.

Suppose now that the contest starts with player 1 ahead, $x_0 > 0$ (the case when $x_0 < 0$ is similar). If both players exert zero effort, player 1 will win and player 2's payoff will be zero. We argue that player 2 cannot do better. Indeed, she would not be willing to spend more than $\frac{V}{2}$ to tie the contest, or more than V to win it with certainty. However, simply by matching player 2's effort starting at a slightly later time, player 1 can tie the contest by spending strictly less than $\frac{V}{2}$, and can win with certainty by spending strictly less than V . Knowing this, player 2 will not exert any effort to begin with.

Next, we consider heterogeneous players. Suppose, without loss, that $c_1 < c_2$, that is, player 1 is the stronger player. The same argument as above can be used to show that she will win the contest with certainty without exerting any effort if $x_0 > 0$. Moreover, if $x_0 = 0$, she can guarantee a win by exerting an infinitesimally small, positive amount of effort. Indeed, player 2 will not want to engage in the race because she is not willing to spend more than $\frac{V}{2}$ to tie, but player 1 can spend strictly less than player 2 and win with certainty. To formalize this outcome, we can use the concept of ϵ -equilibrium (Flesch et al. 2013): fixing an $\epsilon > 0$, an effort level $\delta(\epsilon)$, converging to zero as $\epsilon \rightarrow 0$, can be found such that player 1's payoff from exerting this effort will exceed $V - \epsilon$. Because this holds for any $\epsilon > 0$, effectively we have player 1 winning with zero effort.

Figure C.1. (Color online) Comparison of High-Precision Numerical and Asymptotic Solutions for $q(z)$ 

Notes. (Left) Function $q(z)$ for total expected effort in a symmetric contest with $w_1 = w_2 = 6$. (Right) Function $q(z)$ for total expected effort in an asymmetric contest with $w_1 = 9$, $w_2 = 6$. The solid curves show high-precision numerical solutions to (20), whereas the dots show the asymptotic solutions obtained as described in this section.

The most interesting case is when $x_0 < 0$, that is, the weaker player has a positive head-start. For a large enough $|x_0|$, player 1 will not be able to catch up at a reasonable cost, and player 2 will win with certainty. Yet, when $|x_0|$ is not too large, player 1 should be able to win the contest by exerting effort and moving the state just past $x = 0$. To find the critical value of $|x_0|$ separating the two regimes, because the cost of effort is quadratic, the least costly way of moving the state a given distance is to exert constant effort over the entire duration of the contest. By exerting effort $\bar{\mu}_i$ for all $t \in [0, T]$, player i can move the state by $\bar{\mu}_i T$ at cost $\frac{c_i}{2} \bar{\mu}_i^2 T$. This cost must be below V , implying $\bar{\mu}_i < \sqrt{\frac{2V}{c_i T}}$ for player i to want to exert any effort if it leads to winning with certainty. If both players exert maximum effort, player 1 will be able to catch up and win if $(\sqrt{\frac{2V}{c_1 T}} - \sqrt{\frac{2V}{c_2 T}})T \geq |x_0|$. Therefore, if this condition holds, player 2 will not be willing to engage in the contest and player 1 can win by simply exerting constant effort infinitesimally above $\bar{\mu}_1 = \frac{|x_0|}{T}$. Here, we again need to use ϵ -equilibrium to make the argument precise. In the following proposition, summarizing these results, the equilibrium is understood as ϵ -equilibrium in the limit $\epsilon \rightarrow 0$ if necessary.

Proposition B.2. Suppose $\sigma = 0$ and $c_1 < c_2$, and let $\bar{x} = \sqrt{2VT}(\frac{1}{\sqrt{c_1}} - \frac{1}{\sqrt{c_2}})$. Then there are subgame-perfect equilibria supporting each of the following:

- (i) If $x_0 \geq 0$, both players exert no effort and player 1 wins.
- (ii) If $0 < -x_0 \leq \bar{x}$, player 2 exerts no effort while player 1 exerts constant effort $\frac{|x_0|}{T}$ and wins. Total effort in the contest is $|x_0|$.
- (iii) If $-x_0 > \bar{x}$, both players exert no effort and player 2 wins.

To connect Proposition B.2 to the stochastic case, consider Equation (19). For $x < 0$, $M(x, t) = -x$ is a solution provided $m_1(x, t) = -\frac{x}{T-t}$ and $m_2(x, t) = 0$. Thus, indeed total effort is given by the distance to the origin as in part (ii) of the proposition, and instantaneous effort is constant over time. Let $\hat{x}(t) = x_0 + \bar{\mu}_1 t$ denote the state as a function of time under constant effort $\bar{\mu}_1$. Then

$$m(\hat{x}(t), t) = -\frac{\hat{x}(t)}{T-t} = -\frac{x_0 + \bar{\mu}_1 t}{T-t} = \bar{\mu}_1$$

holds if $\bar{\mu}_1 = -\frac{x_0}{T}$, as in part (ii). It can also be shown directly from (16) that $m_1(x, t) \rightarrow -\frac{x}{T-t}$ and $m_2(x, t) \rightarrow 0$ for $0 < -x \leq \bar{x}$ and $\sigma \rightarrow 0$.

This relatively simple equilibrium analysis of the zero-noise case is possible because the model is in continuous time. It is well known from the analysis of simpler repeated games, such as the prisoner's dilemma (PD), that continuous time can drastically change the structure of equilibria. For example, in a continuous time PD of a finite duration cooperation is sustained as the unique subgame-perfect equilibrium (Simon and Stinchcombe 1989), whereas in discrete time defection is the only equilibrium. The reason is that, in continuous time, to benefit from defection a player needs to be able to defect undetected for some time of a positive measure. When actions are perfectly observable this is impossible because the other player will react immediately; therefore, defection is not profitable and cooperation is sustained. Proposition B.1 for $x_0 = 0$ is similar: The players cooperate by exerting zero effort and reach a Pareto optimal outcome. Similar Pareto optimal outcomes, although asymmetric, are reached with a nonzero head-start or when players are heterogeneous, with the exception of part (ii) of Proposition B.2.

Appendix C. Large Ability Asymptotics

When ability parameters $w_{1,2}$ tend to infinity (for a fixed $\sigma > 0$), either because V goes to infinity or $c_{1,2}$ go to zero, aggregate effort in the contest tends to a constant that only depends on the location of the pivot point z_m . In this section, we characterize the corresponding asymptotic solution to Equation (20).

The main difficulty with solving (20) directly when abilities are large is that coefficient a_0 , Equation (17), diverges, and function $\rho(z)$, Equation (13), becomes close to ± 1 outside a narrow interval around the pivot point z_m . Within that interval, $\rho(z)$ changes sharply from -1 to 1 . As a consequence, functions $A_{\pm}(z)$ are finite outside this interval but become large—of order a_0 —inside of it.

Equation (20), therefore, exhibits a boundary layer type behavior (Verhulst 2005). For concreteness, consider a region $z \geq z_m + \delta$ to the right of the boundary layer, where $\delta > 0$ is chosen so that $\rho(z_m + \delta) \approx 1$. Equation (13) then produces

$$a_0\Phi(z) - \gamma'(\rho_2) = \gamma'(\rho(z)) \approx \frac{1}{1 - \rho(z)^2}, \quad z \geq z_m + \delta,$$

where we used the property that $\ln(1 - \rho) = o\left(\frac{1}{1 - \rho}\right)$ for $\rho \rightarrow 1^-$. This gives, asymptotically,

$$\begin{aligned} A_-(z) &= \frac{a_0}{2\sqrt{2\pi}}[1 - \rho(z)^2] \approx \frac{a_0}{2\sqrt{2\pi}[a_0\Phi(z) - \gamma'(\rho_2)]} \\ &= \frac{1}{2\sqrt{2\pi}[\Phi(z) - \Phi(z_m)]}, \quad z \geq z_m + \delta. \end{aligned}$$

Thus, outside the boundary layer $A_-(z)$ tends to an asymptotic function that depends on parameters only through z_m . Note that $A_+(z) = A_-(z)\rho(z) \approx A_-(z)$ for $z \geq z_m + \delta$. Similarly, to the left of the boundary layer—for $z \leq z_m - \delta$ —we have $\rho(z) \approx -1$ and $A_+(z) = -A_-(z)$. Thus, outside the boundary layer we can define asymptotic functions $A_\pm(z)$ as follows:

$$\begin{aligned} A_\pm^\pm(z) &= \frac{1}{2\sqrt{2\pi}[\Phi(z) - \Phi(z_m)]}, \quad A_\pm^\pm(z) = \text{sgn}(z - z_m)A_\pm^\pm(z), \\ |z - z_m| &\geq \delta. \end{aligned} \quad (\text{C.1})$$

For z outside the boundary layer, functions $A_\pm^\pm(z)$ are well behaved, and Equation (20) can easily be solved. We refer to the corresponding solution, $q_{\text{outer}}(z)$, as the *outer* solution. The challenge is to identify correct boundary conditions for it on either side of z_m . These conditions are found by matching the slopes of the outer solution and the *inner* solution inside the boundary layer, which we identify next.

Inside the boundary layer, preserving only large terms in (20), we have

$$q''(z) + 4A_+(z)e^{-z^2/2}q'(z) + 4A_-(z)e^{-z^2/2} = 0.$$

Integrating once in (z_m, z) , obtain

$$q'(z) = \left[q'(z_m) - 4 \int_{z_m}^z A_-(y)e^{-y^2/2} e^{4 \int_{z_m}^y A_+(s)e^{-s^2/2} ds} \right] e^{-4 \int_{z_m}^z A_+(s)e^{-s^2/2} ds}. \quad (\text{C.2})$$

Recall, from Lemma 4, that $q'(z_0) = 0$ at the unique mode z_0 . When abilities are large, $z_0 \approx z_m$; therefore, we impose the boundary condition $q'(z_m) = 0$. Moreover, from (8) we have

$$\int_y^z A_+(s)e^{-s^2/2} ds = - \int_y^z \frac{A'_-(s) ds}{2A_-(s)} = \frac{1}{2} \ln \frac{A_-(y)}{A_-(z)}.$$

Equation (C.2) then becomes

$$q'(z) = -4A_-(z)^2 \int_{z_m}^z \frac{e^{-s^2/2} ds}{A_-(s)}. \quad (\text{C.3})$$

From (14) and (13), $A_-(z) = \frac{a_0}{2\sqrt{2\pi}}(1 - \rho(z)^2)$ and $\gamma'(\rho(z)) = a_0\Phi(z) - \gamma'(\rho_2)$. This gives $\gamma'(\rho)\rho'(z) = \frac{a_0}{\sqrt{2\pi}}e^{-z^2/2}$, where $\gamma'(\rho) = \frac{2}{(1 - \rho^2)^2}$, and hence $\rho'(z) = \frac{a_0}{2\sqrt{2\pi}}(1 - \rho^2)^2 e^{-z^2/2}$ and (C.3) becomes

$$q'(z) = -4[1 - \rho(z)^2]^2 \int_0^{\rho(z)} \frac{dy}{(1 - y^2)^3}.$$

Using the integral

$$\int \frac{dy}{(1 - y^2)^3} = \frac{y}{4(1 - y^2)^2} + \frac{3y}{8(1 - y^2)} + \frac{3}{16} \ln \frac{1 + y}{1 - y},$$

we finally obtain for the inner solution,

$$q'_{\text{inner}}(z) = -\rho(z) - \frac{3}{2}[1 - \rho(z)^2]\rho(z) - \frac{3}{4}[1 - \rho(z)^2] \ln \frac{1 + \rho(z)}{1 - \rho(z)}. \quad (\text{C.4})$$

To match the inner and outer solutions, for $z \geq z_m + \delta$, $q_{\text{outer}}(z)$ is obtained in the region where $\rho(z) \approx 1$. Considering the limit $\rho(z) \rightarrow 1$ in the inner solution, we obtain at the matching point $q'_{\text{inner}}(z_m + \delta) = -\rho(z_m + \delta) = -1$. Therefore, $q'_{\text{outer}}(z_m + \delta) = -1$ is the boundary condition for the right branch of the outer solution. Thus, we obtain the branch of $q_{\text{outer}}(z)$ for $z \geq z_m + \delta$ by solving (20) in $[z_m + \delta, \infty)$, with $A_\pm(z)$ replaced with $A_\pm^\pm(z)$ from (C.1), and boundary conditions $q'_{\text{outer}}(z_m + \delta) = -1$ and $q_{\text{outer}}(\infty) = 0$. The branch for $z \leq z_m - \delta$ is obtained similarly. There, we solve (20) in $(-\infty, z_m - \delta]$, with boundary conditions $q'_{\text{outer}}(z_m - \delta) = 1$ and $q_{\text{outer}}(-\infty) = 0$. Parameter δ should be chosen so that, on the one hand, it is sufficiently small so that the solutions approach z_m on either side, but at the same time, it is sufficiently large so that $\rho(z_m \pm \delta) \approx \pm 1$. As abilities $w_{1,2}$ increase, the boundary layer shrinks and it becomes easier to choose such a δ .

The method is illustrated in Figure C.1 that shows numerical solutions for the full Equation (20) and asymptotic solutions for relatively large, but not too large, $w_{1,2}$. The approximation works very well even though the full solution can still be obtained. If $w_{1,2}$ are further increased, the direct numerical solution to (20) is no longer available because of singularities, whereas the asymptotic solution is always available. When players are heterogeneous, the asymptotic solution is discontinuous at $z = z_m$. The full solution, albeit continuous, follows that patterns as well: As abilities increase, $q(z)$ is becoming more skewed and its slope at $z = z_m$ increases (Figure 5). In this case, the optimal head-start is close to $z = z_m$, and the pattern of total effort resembles zero-noise behavior. Indeed, when abilities are large, each player exerts effort almost exclusively in her dominance zone. For the stronger player, this includes states when she is behind, between zero and z_m . The difference from the zero-noise case is that when $x = 0$ is surpassed, it still makes sense to exert some effort to avoid losing because of the noise.

Endnotes

¹ The author thanks an anonymous reviewer for this intuition.

² The use of head starts or handicaps as a way to intensify competition when players are heterogeneous has been explored extensively for static contests (Kirkegaard 2012, Franke et al. 2013, Siegel 2014). Handicapping suppliers with lower production costs can also be optimal in a dynamic procurement setting where suppliers become more efficient the more times they were previously selected by the buyer (Lewis and Yildirim 2002). In this case, the buyer can sometimes choose a high-cost supplier to reduce the leader's cost advantage and increase competition. For a review of the literature on various approaches to leveling the playing field in contests, see

Chowdhury et al. (2020). Interestingly, in some cases it may be optimal to bias the contest in favor of the stronger player or even bias a contest between symmetric players (Drugov and Ryvkin 2017, Fu and Wu 2020).

³ For examples of perfectly discriminating contest models, that is, variants of an all-pay auction, see Hillman and Riley (1989), Baye et al. (1996), Moldovanu and Sela (2001), and Siegel (2009).

⁴ The former model can be viewed as a special case of the latter (Jia 2008, Ryvkin and Drugov 2021).

⁵ Rosen (1986), Gradstein and Konrad (1999), Moldovanu and Sela (2006), and Zhang and Wang (2009) provide a discussion of elimination contests.

⁶ See Tsetlin, Gaba and Winkler (2004), Anderson and Cabral (2007), Brookins, Brown and Ryvkin (2016), and Usvitskiy (2020).

⁷ A number of authors investigate optimal information disclosure policies (Halac et al. 2017, Daley and Wang 2018, Bimpikis et al. 2019).

⁸ A tie-breaking rule has no material consequences because $\mathbb{P}(X_T = 0) = 0$.

⁹ We can think of v_i solving (4) in $\mathbb{R} \times [0, T)$ and the boundary and terminal conditions holding in the limits $x \rightarrow \pm\infty$ and $t \rightarrow T^-$, respectively. The values of $v_i(x, T)$ at $x = 0$ can be left undefined because $P(X_T = 0) = 0$ for $\sigma > 0$.

¹⁰ In their setting, monotonicity is possible because the state is bounded.

¹¹ As we show later, up to an affine transformation function $\rho(z)$ is equal to player 1's equilibrium probability of winning the contest in states (x, t) such that $z = \frac{x}{\sigma\sqrt{T-t}}$.

¹² Only one of Conditions (15) can be violated at a time.

¹³ The dependence on time is contained entirely in the factor $\frac{1}{\sqrt{T-t}}$ and in the composite variable z (see Equation (16)); therefore, qualitatively, the effects of abilities at later times are the same as at $t = 0$, except they occur at smaller distances.

¹⁴ Standard numerical methods fail when abilities w_i become large, either because of small noise or because $\frac{V}{c_i}$ are large. The zero-noise limit is discussed in Appendix B. An asymptotic solution for large $\frac{V}{c_i}$ and nonzero noise is discussed in Appendix C.

¹⁵ A nonmonotone dependence of total effort on V and σ was also found for tug-of-war contests by Moscarini and Smith (2011) and Cao (2014).

¹⁶ Throughout this discussion, we assumed that the principal's objective is to maximize total expected effort in the contest, and the prize budget is fixed. In an alternative setting where the prize money, V , is flexible, and the objective is increasing in effort and decreasing in V , there may be even more scope for improvement from splitting the contest because budgets in the smaller subcontests can be optimally adjusted. The author thanks an anonymous reviewer for pointing this out.

¹⁷ For example, Lazear and Oyer (2012, p. 485) write: "An important variable in the Lazear and Rosen (1981) model is the amount of noise—that is, the degree to which luck affects the probability of winning. When there is more noise (so that luck becomes relatively more important and effort relatively less important), workers will try less hard to win, because effort has a reduced effect on whether they win. In production environments that are very uncertain, large raises must be given to offset the tendency by workers to reduce effort."

References

- Anderson A, Cabral LMB (2007) Go for broke or play it safe? Dynamic competition with choice of variance. *RAND J. Econom.* 38(3):593–609.
- Aoyagi M (2010) Information feedback in a dynamic tournament. *Games Econom. Behav.* 70(2):242–260.
- Apestegua J, Palacios-Huerta I (2010) Psychological pressure in competitive environments: Evidence from a randomized natural experiment. *Amer. Econom. Rev.* 100(5):2548–2564.
- Ariely D, Gneezy U, Loewenstein G, Mazar N (2009) Large stakes and big mistakes. *Rev. Econom. Stud.* 76(2):451–469.
- Baye MR, Kovenock D, De Vries CG (1996) The all-pay auction with complete information. *Econom. Theory* 8(2):291–305.
- Benkert J-M, Letina I (2020) Designing dynamic research tournaments. *Amer. Econom. J.: Microeconomics* 12(4):270–289.
- Berger J, Pope D (2011) Can losing lead to winning? *Management Sci.* 57(5):817–827.
- Bimpikis K, Ehsani S, Mostagir M (2019) Designing dynamic contests. *Oper. Res.* 67(2):339–356.
- Brookins P, Brown J, Ryvkin D (2016) Peer information and risk-taking under competitive and non-competitive pay schemes. NBER Working Paper No. w22486, National Bureau of Economic Research, Cambridge, MA.
- Cao D (2014) Racing under uncertainty: Boundary value problem approach. *J. Econom. Theory* 151:508–527.
- Casas-Arce P, Asís Martínez-Jerez F (2009) Relative performance compensation, contests, and dynamic incentives. *Management Sci.* 55(8):1306–1320.
- Chen B, Jiang X, Knyazev D (2016) Head starts and doomed losers: Contest via search. Preprint, submitted November 11, 2017, <https://dx.doi.org/10.2139/ssrn.3067661>.
- Chowdhury SM, Esteve-González P, Mukherjee A (2020) Heterogeneity, leveling the playing field, and affirmative action in contests. Preprint, submitted August 27, 2020, <https://dx.doi.org/10.2139/ssrn.3655727>.
- Daley B, Wang R (2018) When to release feedback in a dynamic tournament. *Decision Anal.* 15(1):11–26.
- Drugov M, Ryvkin D (2017) Biased contests for symmetric players. *Games Econom. Behav.* 103:116–144.
- Ederer F (2010) Feedback and motivation in dynamic tournaments. *J. Econom. Management Strategy* 19(3):733–769.
- Ewerhart C, Teichgräber J (2019) Multi-battle contests, finite automata, and the tug-of-war. Preprint, submitted March 20, 2019, <https://dx.doi.org/10.2139/ssrn.3343709>.
- Flesch J, Kuipers J, Schoenmakers G, Vrieze K (2013) Subgame-perfection in free transition games. *Eur. J. Oper. Res.* 228(1):201–207.
- Franke J, Kanzow C, Leininger W, Schwartz A (2013) Effort maximization in asymmetric contest games with heterogeneous contestants. *Econom. Theory* 52(2):589–630.
- Fu Q, Wu Z (2020) On the optimal design of biased contests. *Theoretical Econom.* 15(4):1435–1470.
- Genakos C, Pagliero M (2012) Interim rank, risk taking, and performance in dynamic tournaments. *J. Political Econom.* 120(4):782–813.
- Gershkov A, Perry M (2009) Tournaments with midterm reviews. *Games Econom. Behav.* 66(1):162–190.
- Gill D, Prowse V (2012) A structural analysis of disappointment aversion in a real effort competition. *Amer. Econom. Rev.* 102(1):469–503.
- Goltsman M, Mukherjee A (2011) Interim performance feedback in multistage tournaments: The optimality of partial disclosure. *J. Labor Econom.* 29(2):229–265.
- Gradstein M, Konrad KA (1999) Orchestrating rent seeking contests. *Econom. J. (London)* 109(458):536–545.
- Halac M, Kartik N, Liu Q (2017) Contests for experimentation. *J. Political Econom.* 125(5):1523–1569.
- Harris C, Vickers J (1987) Racing with uncertainty. *Rev. Econom. Stud.* 54(1):1–21.
- Hillman AL, Riley JG (1989) Politically contestable rents and transfers. *Econom. Politics* 1(1):17–39.
- Iqbal H, Krumer A (2019) Discouragement effect and intermediate prizes in multi-stage contests: Evidence from Davis Cup. *Eur. Econom. Rev.* 118:364–381.

- Jia H (2008) A stochastic derivation of the ratio form of contest success functions. *Public Choice* 135(3-4):125–130.
- Kirkegaard R (2012) Favoritism in asymmetric contests: Head starts and handicaps. *Games Econom. Behav.* 76(1):226–248.
- Kiselev A, Roquejoffre J-M, Ryzhik L (2017) Appetizers in Nonlinear PDE. Working paper, Stanford University, Stanford, CA.
- Klump T, Polborn MK (2006) Primaries and the New Hampshire effect. *J. Public Econom.* 90(6-7):1073–1114.
- Konrad KA, Kovenock D (2009) Multi-battle contests. *Games Econom. Behav.* 66:256–274.
- Lang M, Seel C, Strack P (2014) Deadlines in stochastic contests. *J. Math. Econom.* 52:134–142.
- Lazear EP, Oyer P (2012) Personnel economics. *The Handbook of Organizational Economics* (Princeton University Press, Princeton, NJ), 479–519.
- Lazear EP, Rosen S (1981) Rank-order tournaments as optimum labor contracts. *J. Political Econom.* 89(5):841–864.
- Lewis TR, Yildirim H (2002) Managing dynamic competition. *Amer. Econom. Rev.* 92(4):779–797.
- Moldovanu B, Sela A (2001) The optimal allocation of prizes in contests. *Amer. Econom. Rev.* 91(3):542–558.
- Moldovanu B, Sela A (2006) Contest architecture. *J. Econom. Theory* 126(1):70–96.
- Morgan J, Tumlinson J, Vardy FJ (2019) The limits of meritocracy. Preprint, submitted November 29, 2021, <https://dx.doi.org/10.2139/ssrn.3216950>.
- Moscarini G, Smith L (2011) Optimal dynamic contests. Working paper, Yale University, New Haven, CT.
- Ridlon R, Shin J (2013) Favoring the winner or loser in repeated contests. *Marketing Sci.* 32(5):768–785.
- Rosen S (1986) Prizes and incentives in elimination tournaments. *Amer. Econom. Rev.* 76(4):701–715.
- Ryvkin D (2011) Fatigue in dynamic tournaments. *J. Econom. Management Strategy* 20(4):1011–1041.
- Ryvkin D, Drugov M (2021) The shape of luck and competition in winner-take-all tournaments. *Theoretical Econom.* 15(4):1587–1626.
- Seel C, Strack P (2013) Gambling in contests. *J. Econom. Theory* 148(5):2033–2048.
- Seel C, Strack P (2016) Continuous time contests with private information. *Math. Oper. Res.* 41(3):1093–1107.
- Siegel R (2009) All-pay contests. *Econometrica* 77(1):71–92.
- Siegel R (2014) Asymmetric contests with head starts and nonmonotonic costs. *Amer. Econom. J. Microeconom.* 6(3):59–105.
- Simon LK, Stinchcombe MB (1989) Extensive form games in continuous time: Pure strategies. *Econometrica* 57(5):1171–1214.
- Taylor CR (1995) Digging for golden carrots: An analysis of research tournaments. *Amer. Econom. Rev.* 85(4):872–890.
- Tsetlin I, Gaba A, Winkler RL (2004) Strategic choice of variability in multiround contests and contests with handicaps. *J. Risk Uncertainty* 29(2):143–158.
- Tullock G (1980) Efficient rent seeking. Buchanan JM, Tollison RD, Tullock G, eds. *Toward a Theory of the Rent-Seeking Society* (Texas A&M University Press, College Station), 97–112.
- Usvitskiy A (2020) Risk-taking in dynamic contests. Working paper, The School of Advanced Studies, Tyumen, Russia.
- Verhulst F (2005) Methods and Applications of Singular Perturbations: Boundary Layers and Multiple Timescale Dynamics, vol. 50 of Texts in Applied Mathematics (Springer, Berlin).
- Wang R (2017) Essays on dynamic tournaments. PhD dissertation, Duke University, Chapel Hill, NC.
- Yildirim H (2005) Contests with multiple rounds. *Games Econom. Behav.* 51(1):213–227.
- Zhang J, Wang R (2009) The role of information revelation in elimination contests. *Econom. J. (London)* 119(536):613–641.